

SGA 5 – Exposé I and Exposé III

Dualizing Complexes, continued; Lefschetz Formula and local terms, continued

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Working English LaTeX translation

Exposé I. Dualizing Complexes, continued

3. Existence of dualizing complexes

3.1. Preliminaries

As was said in the introduction, the object of this section is to prove that, on a “good” regular prescheme X , the sheaf A_X is dualizing, where n is assumed prime to the residual characteristics.

In fact, one would like this to hold on excellent regular preschemes of equal characteristic; but there is no doubt that the proof given below will extend to them once resolution of singularities and the purity theorem are available in that case.

We first present the various ingredients which will be used.

3.1.1. Singular locus. Let X be a prescheme, locally noetherian. We shall say that X satisfies condition *(reg)* if the following equivalent properties hold (EGA IV, 6.12.3):

- (i) for every sub-prescheme Y of X , the set $\text{Reg}(Y)$ of regular points of Y is open in Y ;
- (ii) for every closed integral sub-prescheme Y of X , the set $\text{Reg}(Y)$ contains a non-empty open subset of Y .

The condition *(reg)* is satisfied if X is locally of finite type over a complete noetherian local ring, for example over a field (EGA IV, 6.12.8), or over a Dedekind ring whose field of fractions has characteristic zero, for example over $\mathbb{Z}$ (EGA IV, 6.12.6); more generally, it is satisfied if X is excellent (EGA IV, 7.8.6).

3.1.2. Dévissage of constructible sheaves. Let X be a noetherian prescheme, and let $\mathcal{C}$ be a strictly full subcategory of the category $\text{Cons}(X)$ of constructible A_X -modules. Suppose that $\mathcal{C}$ is stable under direct factors and extensions. The following conditions are equivalent:

- (i) $\mathcal{C} = \text{Cons}(X)$;

- (ii) C contains sheaves of the form $i_!(F)$, where $i : Y \rightarrow X$ is the immersion of an integral sub-prescheme and F is locally constant on Y ;
- (iii) C contains sheaves of the form $f_* i_!(A'_U)$, where $i : U \rightarrow Y$ is an open subset of an integral prescheme, $f : Y \rightarrow X$ is a finite morphism such that the generic point of Y is separable over its image, and $A' = \mathbb{Z}/\ell\mathbb{Z}$, for ℓ a prime divisor of n .

If X satisfies (reg) (3.1.1), these conditions are again equivalent to the following:

- (i bis) the analogue of (ii), with Y moreover regular;
- (ii bis) the analogue of (iii), with U regular.

If, in addition, X is affine, then in (ii), respectively in (ii bis), one may suppose Y closed in an open subset of the form $D(f)$, with $f \in \Gamma(X, \mathcal{O}_X)$.

Proof. The equivalence of (i)–(iii) was proved in SGA A, Exposé IX, 2.5 and 5.8. The remaining assertions follow by noetherian induction. $\square$

3.1.3. Cohomological dimension. Let X be a locally noetherian prescheme. We denote by $(\text{cdloc})_n$ the following condition. For every sub-prescheme Y of X , every geometric point $\bar{y}$ of Y , and every

$$f \in \Gamma(\tilde{Y}(\bar{y}), \mathcal{O}_{\tilde{Y}, \bar{y}}),$$

where $\tilde{Y}(\bar{y})$ denotes the strict localization of Y at $\bar{y}$ (SGA A, Exposé VIII, 7.1), one has

$$\text{cd}_{\mathbf{n}}(\tilde{Y}(\bar{y}) - V(f)) \leq \dim \mathcal{O}_{Y, \bar{y}} \quad (= \dim \mathcal{O}_{Y, y} \text{ by EGA IV, 6.1.3}),$$

where $\mathbf{n}$ is the set of prime divisors of n .

The condition $(\text{cdloc})_n$ is satisfied if X is locally of finite type over a field (SGA A, Exposé XIV, 3.5), or is an excellent prescheme of characteristic zero (SGA A, Exposé XIX, 6.2).¹

3.1.4. Absolute cohomological purity. Let X be a regular prescheme, and let Y be a regular closed sub-prescheme of X , of codimension d at every point. One says that the pair (X, Y) satisfies the absolute cohomological purity theorem, relative to n , if

$$\underline{H}_Y^i(A_X) = 0 \quad (i \neq 2d),$$

and if the “local fundamental class” homomorphism (SGA 4^{1/2}, Cycle 2.2)

$$A_Y \longrightarrow \underline{H}_Y^{2d}((\mu_n)_X^{\otimes d})$$

¹Added in 1977: or, by a recent result of G. Ofer, if X is excellent, regular, and of dimension 2.

is an isomorphism.

One says that the absolute cohomological purity theorem is true on X if every pair (U, Y) as above satisfies it, where U is an open subset of X .

Absolute purity is true on X if X is smooth over a perfect field of characteristic prime to n (SGA A, Exposé XVI, 3.9), or if X is an excellent prescheme of characteristic zero (SGA A, Exposé XIX, 3.2), or if X is regular of dimension 1 (cf. 5.1).²

3.1.5. Hironaka-style resolution of singularities.

- a) Let X be a regular prescheme, and let $D \geq 0$ be a divisor on X . One says that D has *strict normal crossings* if there exists a finite family $(f_i)_{i \in I}$ of elements of $\Gamma^*(X, \mathcal{O}_X)$ such that

$$D = \sum_{i \in I} \text{Div}(f_i)$$

(EGA IV, 21), and such that, for every $x \in \text{Supp}(D)$, the germs $(f_i)_x$ which lie in $\mathfrak{m}_x$ form part of a regular system of parameters. One says that D has *normal crossings* if, locally for the étale topology, D has strict normal crossings.

The standard example of a normal crossings divisor is furnished by a locally finite union of coordinate hyperplanes in affine space.

- b) Let X be a locally noetherian prescheme. We shall say that X is *strongly desingularizable* if the following condition is satisfied: for every finite morphism $f : Y \rightarrow X$, where Y is integral and its generic point is separable over its image, and every regular open subset U of Y , there exists a proper birational morphism $h : Y' \rightarrow Y$, with Y' regular, such that h induces an isomorphism from $U' = h^{-1}(U)$ onto U , and $Y' - U'$ is a normal crossings divisor.

Here are cases where X is strongly desingularizable:

- a) X is excellent of characteristic zero;
- b) X is locally noetherian, regular, and of dimension ≤ 1 ;
- c) X is of finite type over $\mathbb{Z}$ or over a perfect field, and of dimension ≤ 2 .³

3.1.6. Cohomological finiteness. Let $f : X \rightarrow Y$ be a morphism of preschemes, and let $F \in \text{Ob } D_c^+(X)$. One has $Rf_*(F) \in \text{Ob } D_c^+(Y)$ in the following cases:

²Added in 1977: or, by a recent result of G. Ofer, if X is excellent, regular, and of dimension 2.

³Added in 1977: more generally, if X is excellent noetherian of dimension ≤ 2 , by H. Hironaka, *Desingularization of excellent surfaces*, notes by B. Bennett, Bowdoin College, 1967, to be supplemented if necessary by H. Hironaka, *Certain numerical characters of singularities*, J. Math. Kyoto Univ. 10 (1970), 151–187.

- a) f is proper and of finite presentation (SGA A, Exposé XIV, 1.1);
- b) “when resolution of singularities and the purity theorem are available”, in particular:
 - (i) if X and Y are locally of finite type over a field k of characteristic zero and f is a k -morphism of finite type (SGA A, Exposé XVI, 5.1);
 - (ii) if Y is excellent of characteristic zero and f is of finite type (SGA A, Exposé XIX, 5.1);
 - (iii) if X is locally of finite type over a field and $\dim(X) \leq 2$ (SGA A, Exposé XIX, 5.1).

3.2. The quasi-injective dimension of A_X

Proposition 3.2.1. Let X be a regular prescheme whose residual characteristics are prime to n . Suppose that X satisfies (reg) (3.1.1) and (cdloc) $_n$ (3.1.3), and that absolute purity is true on X (3.1.4). Then, for every constructible A_X -module F and every $x \in X$, one has

$$\underline{\mathrm{Ext}}^i(F, A_X)_x = 0 \quad \text{for } i > 2 \dim \mathcal{O}_{X,x}. \quad (*)$$

Proof. The question being local, we may suppose X affine. By dévissage (3.1.2), it is enough to verify (*) for $F = i_!(G)$, where $i : Y \rightarrow X$ is the immersion of a regular integral subscheme, closed in an open subset $D(f)$, $f \in A(X)$, and G is a locally constant constructible sheaf on Y . We must compute

$$\underline{\mathrm{Ext}}^q(F, A_X) = H^q(\mathrm{R} \underline{\mathrm{Hom}}(F, A_X)).$$

The projection formula for i (SGA A, Exposé XVIII) gives

$$\mathrm{R} \underline{\mathrm{Hom}}(i_!G, A_X) \xrightarrow{\sim} \mathrm{R} i_* \mathrm{R} \underline{\mathrm{Hom}}(G, \mathrm{R}^!i(A_X)).$$

By purity,

$$\mathrm{R}^!i(A_X) \xrightarrow{\sim} (\mu_n)_Y^{\otimes d}[-2d],$$

where $d = \mathrm{codim}(Y, X)$. Put $T = (\mu_n)_Y^{\otimes d}$. Since G is locally constant constructible, for every geometric point $\bar{y}$ of Y and every p one has

$$\underline{\mathrm{Ext}}^p(G, T)_{\bar{y}} \xrightarrow{\sim} \mathrm{Ext}^p(G_{\bar{y}}, T_{\bar{y}}).$$

But $T_{\bar{y}}$ is injective, being isomorphic to $A = \mathbb{Z}/n\mathbb{Z}$. Hence $\underline{\mathrm{Ext}}^p(G, T) = 0$ for $p \neq 0$, and

$$\mathrm{R} \underline{\mathrm{Hom}}(G, T) \xrightarrow{\sim} \underline{\mathrm{Hom}}(G, T).$$

Thus

$$R\mathbf{Hom}(F, A_X) \xrightarrow{\sim} Ri_*(\mathcal{H})[-2d], \quad \mathcal{H} = \mathbf{Hom}(G, T).$$

Let Z be the schematic closure of Y in X (EGA I, 9.5.10). Then $Y = Z \cap D(f)$ (EGA 0_I, 2.1.6). Denote by $j : Y \rightarrow Z$ and $k : Z \rightarrow X$ the canonical immersions. We have $Ri_* = k_*Rj_*$, so we are reduced to computing $Rj_*(\mathcal{H})$. If $\bar{x}$ is a geometric point of Z , let $\tilde{Z} = \text{Spec}(\mathcal{O}_{Z, \bar{x}})$ be the strict localization, let $\tilde{f}$ be the image of f in $\mathcal{O}_{Z, \bar{x}}$, let $\tilde{Y} = \tilde{Z} \times_Z Y = \tilde{Z} - V(\tilde{f})$, and let $\tilde{\mathcal{H}}$ be the inverse image of $\mathcal{H}$ on $\tilde{Y}$. By SGA A, Exposé VIII, 5.2,

$$R^q j_*(\mathcal{H})_{\bar{x}} \xrightarrow{\sim} H^q(\tilde{Y}, \tilde{\mathcal{H}}),$$

hence

$$\mathbf{Ext}^q(F, A_X)_{\bar{x}} \xrightarrow{\sim} H^{q-2d}(\tilde{Y}, \tilde{\mathcal{H}}).$$

The condition $(\text{cdloc})_n$ implies that this group is zero for

$$q - 2d > \dim \mathcal{O}_{Z, \bar{x}}.$$

Using

$$d = \text{codim}_{\bar{x}}(Z, X) \leq \dim \mathcal{O}_{X, \bar{x}}$$

(EGA IV, 5.1.3) and

$$\dim \mathcal{O}_{Z, \bar{x}} = \dim \mathcal{O}_{X, \bar{x}} - \text{codim}_{\bar{x}}(Z, X)$$

(EGA 0_{IV}, 16.5.12 and EGA IV, 5.1.9), one obtains

$$\mathbf{Ext}^q(F, A_X)_x = 0 \quad (q > 2 \dim \mathcal{O}_{X, x}),$$

as required. □

Remark 3.2.2. The bound obtained in 3.2.1 is the best possible. Indeed, let x be a point of X , and let Y be the closed integral sub-prescheme of X whose generic point is x . Since X satisfies (reg), there exists a regular open subset U of Y containing x . By purity,

$$\mathbf{Ext}^{2d}(A_U, A_X)|_U = (\mu_n)_U^{\otimes d},$$

where $d = \dim \mathcal{O}_{X, x} = \text{codim}(Y, X)$.

Corollary 3.2.3. Let X be a regular prescheme of dimension d . If X is smooth over a field k of characteristic prime to n , or if X is excellent of characteristic zero, then X satisfies the conclusion of 3.2.1; consequently

$$\dim . q. \text{inj}(A_X) \leq 2d.$$

Proof. If X is excellent of characteristic zero, or if k is perfect of characteristic prime to n and

X is smooth over k , then the hypotheses of 3.2.1 are satisfied by 3.1.1, 3.1.3, and 3.1.4. The conclusion is still valid when X is smooth over an arbitrary field k , as follows immediately after the base change $\mathrm{Spec}(k') \rightarrow \mathrm{Spec}(k)$, where k' is a perfect closure of k (SGA A, Exposé VIII, 1.1). $\square$

Lemma 3.2.4. Let X be a prescheme satisfying the conclusion of 3.2.1 and of dimension $\leq d$. If there exists an integer N such that X has cohomological dimension $\leq N$, then

$$\dim .inj(A_X) \leq 2d + N.$$

If, moreover, for every closed subset Y of X , the implication

$$\mathrm{codim}(Y, X) \geq p \quad \Rightarrow \quad \mathrm{cd}_{\mathbf{n}}(Y) \leq N - 2p$$

holds, then

$$\dim .inj(A_X) \leq N.$$

Proof. The first assertion follows immediately from 1.5. For the second, one argues as in 1.5. We are reduced to proving, for F constructible on X , the relation

$$\mathrm{Ext}^i(X; F, A_X) = 0 \quad (i > N).$$

For this, inspect the spectral sequence

$$E_2^{pq} = H^p(X; \underline{\mathrm{Ext}}^q(F, A_X)) \Rightarrow \mathrm{Ext}^*(X; F, A_X).$$

By 3.2.1, the sheaves $\underline{\mathrm{Ext}}^{2i}(F, A_X)$ and $\underline{\mathrm{Ext}}^{2i-1}(F, A_X)$ are concentrated in codimension $\geq i$. Therefore $E_2^{p,2i} = E_2^{p,2i-1} = 0$ for $p > N - 2i$, by the hypothesis. Hence $E_2^{pq} = 0$ for $p + q > N$, and the assertion follows. $\square$

Corollary 3.2.5. If X is a smooth prescheme of dimension d over a separably closed field k of characteristic prime to n , then

$$\dim .inj(A_X) = 2d.$$

Proof. By 3.2.3 and SGA A, Exposé X, 4.2, we are in the situation of 3.2.4. The hypotheses of SGA A, Exposé X, 4.2 are satisfied by SGA A, Exposé X, 2.1 and 3.2; hence $\dim .inj(A_X) \leq 2d$. The equality follows from purity: if x is a closed point of codimension d in X , then

$$\mathrm{Ext}^{2d}(X, A_x, A_X) = H_x^{2d}(A_X) \xrightarrow{\sim} (\mu_n)^{\otimes d}(k) \neq 0.$$

$\square$

3.3. Constructibility of the sheaves $\underline{\mathrm{Ext}}^i(F, A_X)$

Proposition 3.3.1. Let X be a prescheme.

a) The following conditions are equivalent:

(i) for $F \in \mathrm{Ob} D_c^-(X)$ and $G \in \mathrm{Ob} D_c^+(X)$, one has

$$\mathrm{R} \underline{\mathrm{Hom}}(F, G) \in \mathrm{Ob} D_c^+(X);$$

(i bis) for every pair (F, G) of constructible A_X -modules, one has $\mathrm{R} \underline{\mathrm{Hom}}(F, G) \in \mathrm{Ob} D_c^+(X)$, i.e. the sheaves $\underline{\mathrm{Ext}}^i(F, G)$ are constructible for all i ;

(ii) for every open immersion $i : U \rightarrow X$ and every constructible A_U -module F , the sheaves $\mathrm{R}^p i_*(F)$ are constructible for all p ;

(ii bis) for every immersion $i : Y \rightarrow X$ and every constructible A_Y -module F , the sheaves $\mathrm{R}^p i_*(F)$ are constructible for all p ;

(ii ter) for every immersion $i : Y \rightarrow X$ and every locally constant constructible A_Y -module F , the sheaves $\mathrm{R}^p i_*(F)$ are constructible for all p ;

(iii) for every finite morphism, respectively every closed immersion, $f : Y \rightarrow X$, and every $F \in \mathrm{Ob} D_c^+(X)$, one has $\mathrm{R}^! f(F) \in \mathrm{Ob} D_c^+(Y)$.

b) These conditions are satisfied in each of the following two cases:

(i) $\dim(X) \leq 1$;

(ii) resolution of singularities and purity are available on X in the sense of 3.1.6(b).

Proof. The equivalence of (i) and (i bis) follows from the standard spectral sequence passing from sheaf-Ext to hyper-Ext.

(i bis) $\Rightarrow$ (ii). Let $\overline{F}$ be a constructible A_X -module extending F , for example $\overline{F} = i_! F$. By duality (SGA A, Exposé XVIII, 3.1.10),

$$\mathrm{R} i_* F = \mathrm{R} i_* \mathrm{R} \underline{\mathrm{Hom}}(A_U, i^! \overline{F}) \xrightarrow{\sim} \mathrm{R} \underline{\mathrm{Hom}}(i_! A_U, \overline{F}),$$

and this proves the claim.

(ii) $\Rightarrow$ (ii bis). Factor i as a closed immersion followed by an open immersion.

(ii bis) $\Rightarrow$ (i bis). The question is local, so we may suppose X affine. By the usual dévissage (SGA A, Exposé IX, 2.5), it is enough to verify (ii) for $F = i_! M$, where $i : Y \rightarrow X$ is the immersion of a sub-prescheme and M is a locally constant constructible A_Y -module. By the projection formula (SGA A, Exposé XVIII),

$$\mathrm{R} \underline{\mathrm{Hom}}(i_! M, G) \xrightarrow{\sim} \mathrm{R} i_* \mathrm{R} \underline{\mathrm{Hom}}(M, \mathrm{R} i^! G),$$

and by a spectral sequence we are reduced to proving that

$$\mathrm{R}\underline{\mathrm{Hom}}(M, \mathrm{R}i^!G) \in \mathrm{Ob}\, \mathrm{D}_c^+(Y).$$

First prove $\mathrm{R}i^!G \in \mathrm{Ob}\, \mathrm{D}_c^+(Y)$. After factoring i , we may suppose that Y is closed. Let $j : U = X - Y$ be the complementary open immersion. Applying $\mathrm{R}\underline{\mathrm{Hom}}(\ , G)$ to

$$0 \longrightarrow A_{U,X} \longrightarrow A_X \longrightarrow A_Y \longrightarrow 0$$

gives a distinguished triangle. Since $\mathrm{R}\underline{\mathrm{Hom}}(A_X, G) = G$, and

$$\mathrm{R}\underline{\mathrm{Hom}}(A_{U,X}, G) \xrightarrow{\sim} \mathrm{R}j_*j^!G$$

belongs to $\mathrm{D}_c^+(X)$ by hypothesis, it follows that

$$\mathrm{R}\underline{\mathrm{Hom}}(A_Y, G) \xrightarrow{\sim} i_*\mathrm{R}i^!G$$

belongs to $\mathrm{D}_c^+(X)$. Resolving M locally on Y , for the étale topology, by free A_Y -modules, another dévissage gives

$$\mathrm{R}\underline{\mathrm{Hom}}(M, \mathrm{R}i^!G) \in \mathrm{Ob}\, \mathrm{D}_c^+(Y).$$

This proves the implication.

(ii bis) $\Rightarrow$ (ii ter) is trivial.

(ii ter) $\Rightarrow$ (ii). Let $i : U \rightarrow X$ be an open immersion and let F be a constructible A_U -module. We must prove $\mathrm{R}i_*F \in \mathrm{Ob}\, \mathrm{D}_c^+(X)$. The question being local, suppose X noetherian and argue by noetherian induction on X . We may suppose $U \neq \emptyset$. Then there exists a non-empty open subset $j : V \rightarrow U$ such that j^*F is locally constant (SGA A, Exposé IX, 2.4). Let M be the mapping cylinder of

$$F \longrightarrow \mathrm{R}j_*j^*F.$$

Applying $\mathrm{R}i_*$ gives a distinguished triangle

$$\mathrm{R}i_*F \longrightarrow \mathrm{R}i_*\mathrm{R}j_*j^*F \longrightarrow \mathrm{R}i_*M \xrightarrow{[1]}.$$

Condition (ii ter) implies

$$\mathrm{R}i_*\mathrm{R}j_*j^*F \xrightarrow{\sim} \mathrm{R}(ij)_*j^*F \in \mathrm{Ob}\, \mathrm{D}_c^+(X).$$

Consequently,

$$\mathrm{R}j_*j^*F \xrightarrow{\sim} i^*\mathrm{R}(ij)_*j^*F \in \mathrm{Ob}\, \mathrm{D}_c^+(U).$$

Thus $M \in \mathrm{Ob}\, \mathrm{D}_c^+(U)$, since $F \in \mathrm{Ob}\, \mathrm{D}_c^+(U)$. Moreover, the cohomology of M is supported on a closed subset Z of U distinct from U . Applying the induction hypothesis to $U \cap \overline{Z} \rightarrow \overline{Z}$,

we deduce $Ri_*M \in \text{Ob } D_c^+(X)$, and hence $Ri_*F \in \text{Ob } D_c^+(X)$.

The equivalence of (i)–(ii ter) with condition (iii) is left to the reader; use 1.14.

For b)(i), normalization reduces one to the case where X is regular, and one uses purity, which will be proved in 5.1. For b)(ii), see 3.1.6(b). $\square$

3.4. The biduality theorem on regular preschemes

Theorem 3.4.1, cf. [H], V, 2.2. Let X be a regular locally noetherian prescheme of finite dimension. Suppose that X is strongly desingularizable (3.1.5), satisfies (reg) (3.1.1) and (cdloc) $_n$ (3.1.3), and that absolute purity (3.1.4) is true on every prescheme smooth over X . Then A_X is dualizing.

Proof. We must prove that A_X satisfies conditions (i)–(iii) of 1.6. Condition (i) follows from 3.2.1. Condition (ii) follows from 3.3.1(b). It remains to prove condition (iii). Thus, by 1.7, for every constructible A_X -module F we must prove that the canonical homomorphism

$$F \longrightarrow \underline{D}_X \underline{D}_X F, \quad \underline{D}_X = R \underline{\text{Hom}}(_, A_X),$$

is an isomorphism. The question being local, suppose X noetherian. By dévissage (3.1.2, iii bis), it is enough to treat sheaves

$$F = f_* i_!(A'_U),$$

where $i : U \rightarrow Y$ is the inclusion of a regular open subset of an integral prescheme, $f : Y \rightarrow X$ is finite and the generic point of Y is separable over its image, and $A' = \mathbb{Z}/\ell\mathbb{Z}$ for a prime divisor ℓ of n .

Because X is strongly desingularizable, there is a proper birational morphism $h : Y' \rightarrow Y$, with Y' regular, inducing an isomorphism from $U' = h^{-1}U$ to U , and such that $Y' - U'$ is a normal crossings divisor. By base change (SGA A, Exposé XVII),

$$i_!(A'_U) \xrightarrow{\sim} Rh_* i'_!(A'_{U'}),$$

where $i' : U' \rightarrow Y'$ is the inclusion. Hence, with $g = fh$,

$$f_* i_!(A'_U) \xrightarrow{\sim} f_* Rh_* i'_!(A'_{U'}) \xrightarrow{\sim} Rg_* i'_!(A'_{U'}).$$

By 1.11, it suffices to prove that the pair

$$(i'_!(A'_{U'}), R^!g(A_X))$$

is bidualizing. Locally immersing Y' in a prescheme X' smooth over X , and applying purity on X' , one sees that $R^!g(A_X)$ is locally isomorphic to $A_{Y'}$, up to a translation of degrees. We are therefore reduced to the following special case.

Lemma 3.4.2. Let X be a locally noetherian regular prescheme on which absolute purity is true. Let $Y = Y_1 \cup \cdots \cup Y_p$ be a strict normal crossings divisor on X , and put $U = X - Y$. Then the pair $(A'_{U,X}, A_X)$ is bidualizing.

Proof. By the exact sequence

$$0 \longrightarrow A'_{U,X} \longrightarrow A'_X \longrightarrow A'_Y \longrightarrow 0,$$

it is equivalent to verify that (A'_Y, A_X) is bidualizing. We argue by induction on p .

If $p = 1$, let $i : Y \rightarrow X$ be the immersion. By 1.11, it is enough to verify that

$$(A'_Y, R^!i(A_X))$$

is bidualizing. By purity,

$$R^!i(A_X) \xrightarrow{\sim} (\mu_n)_{Y'}^{\otimes 1}[-2].$$

Thus the canonical homomorphism $A'_Y \rightarrow \underline{D}_Y \underline{D}_Y(A'_Y)$ is an isomorphism, as is checked trivially fibre by fibre, using Pontryagin duality for A -modules.

Suppose $p \geq 2$, and put $Y' = Y_1 \cup \cdots \cup Y_{p-1}$. We have an exact sequence

$$0 \longrightarrow A'_{(Y-Y'),X} \longrightarrow A'_Y \longrightarrow A'_{Y'} \longrightarrow 0.$$

By the induction hypothesis, $(A'_{Y'}, A_X)$ is bidualizing. By the five lemma, we are reduced to biduality for $(A'_{(Y-Y'),X}, A_X)$. But again there is an exact sequence

$$0 \longrightarrow A'_{(Y-Y'),X} \longrightarrow A'_{Y_p} \longrightarrow A'_{Y_p \cap Y'} \longrightarrow 0.$$

Since (A'_{Y_p}, A_X) is bidualizing, it remains to prove the same for $(A'_{Y_p \cap Y'}, A_X)$. Applying 1.11 again and using purity, we are finally reduced to biduality for this pair on Y_p , where the induction hypothesis applies. $\square$

This completes the proof of Theorem 3.4.1.

Corollary 3.4.3. Let S be an excellent noetherian regular prescheme of characteristic zero. Then A_S is dualizing. Moreover, if $f : X \rightarrow S$ is a morphism of finite type, then $R^!f(A_S)$ is dualizing.

Proof. The first assertion follows because S satisfies the hypotheses of 3.4.1, as recalled in 3.1. For the second, one must first verify that $R^!f(A_S)$ has finite quasi-injective dimension. Cover S by finitely many affine open subsets S_i such that $X_i = f^{-1}(S_i)$ is a finite union of affine open subsets X_{ij} immersed as closed subschemes in preschemes X'_{ij} smooth of finite dimension over S_i . From 3.2.3 and 1.6 it follows that

$$\dim . q. inj(R^!f(A_S)) \leq 2 \sup \dim X'_{ij}.$$

It remains to verify conditions (ii) and (iii) of 1.7. Since these are local on X for the étale topology, we may suppose that f factors as $f = f' i$, where $f' : X' \rightarrow S$ is smooth of relative dimension d , and $i : X \rightarrow X'$ is a closed immersion. Then

$$R^! f(A_S) \simeq R^! i R^! f'(A_S).$$

But $R^! f'(A_S)$ is locally isomorphic to $A_{X'}$, up to the translation by $2d$, hence is dualizing by the first assertion. By 1.15, $R^! f(A_S)$ is dualizing, and in particular satisfies conditions (ii) and (iii) of 1.7. $\square$

Corollary 3.4.4. Let k be a field, and let $f : X \rightarrow S = \operatorname{Spec}(k)$ be a morphism locally of finite type, with $\dim(X) \leq 2$. Then $R^! f(A_S)$ is dualizing.

Proof. Replacing k by its perfect closure, which is harmless for the étale topology (SGA A, Exposé VIII, 1.1), we may suppose k perfect. Cover X by affine open subsets X_i of finite type over k . Since $\dim(X_i) \leq 2$, the normalization lemma makes X_i finite over a scheme X'_i smooth over S and of dimension ≤ 2 . By 1.15 this reduces us to the case where f is smooth, hence locally

$$f^!(A_S) \simeq A_X[2d].$$

Thus it remains to show that A_X is dualizing in this case; this is exactly 3.4.1, since the hypotheses are satisfied (see 3.1). $\square$

Remark 3.4.5. We shall see below that if X is regular of dimension ≤ 1 , then A_X is dualizing.

4. Local duality

The purpose of this section is to present the notion of biduality developed in the preceding sections in a form closer to geometric intuition.

4.1

Let X be a prescheme, let $i : x \rightarrow X$ be a closed point with separably closed residue field, and let K_X be a dualizing complex on X (1.7). By 1.15, $K_{\{x\}} = R^! i(K_X)$ is a dualizing complex on x , hence by 2.1 and 3.4.4 is isomorphic to $A[r]$ for some $r \in \mathbb{Z}$, if A is local. We shall say that K_X is *normalized at x* if $r = 0$ and if an isomorphism $K_{\{x\}} \xrightarrow{\sim} A$ has been chosen.

For example, if X is smooth of relative dimension d over the spectrum of a separably closed field, the complex $(\mu_n)_X^{\otimes d}[2d]$ is, by purity (SGA A, Exposé XVI, 3; SGA 4^{1/2}, Cycle 2.3), canonically normalized at every closed point of X .

4.2

Let X be a prescheme, let $i : Y \rightarrow X$ be a closed sub-prescheme, and let K_X be a dualizing complex on X . Then $K_Y = R^!i(K_X)$ is a dualizing complex on Y (1.15), and one has the complementary induction formula (1.12 b)(ii)): for $F \in \text{Ob } D_c^b(X)$,

$$i^* \underline{D}_X(F) \xrightarrow{\sim} \underline{D}_Y R^!i(F), \quad (4.2.1)$$

where

$$\underline{D}_X = R \underline{\text{Hom}}(_, K_X), \quad \underline{D}_Y = R \underline{\text{Hom}}(_, K_Y).$$

In particular, take $Y = \{x\}$, where x is a closed point of X with separably closed residue field, and suppose K_X normalized at x . Since A_x is injective, (4.2.1) becomes

$$\underline{D}_X(F)_x \xrightarrow{\sim} \underline{\text{Hom}}^*(R^!i(F), A). \quad (4.2.2)$$

Since $R^!i(F)$ is also $R\Gamma_x(F)$ by definition, one obtains a perfect pairing in $D(A)$,

$$\underline{D}_X(F)_x \times R\Gamma_x(F) \longrightarrow A, \quad (4.2.2')$$

which gives perfect pairings between finite groups

$$\underline{H}^i(\underline{D}_X F)_x \times \underline{H}_x^{-i}(F) \longrightarrow A. \quad (4.2.2'')$$

For example, if X is smooth of dimension d over a separably closed field and $x \in X$ is closed, then, for $F \in \text{Ob } D_c^b(X)$, there is a canonical perfect duality between finite groups

$$\underline{\text{Ext}}^{2d-i}\left(F, (\mu_n)_X^{\otimes d}\right)_x \times \underline{H}_x^i(F) \longrightarrow A. \quad (4.2.2''')$$

4.3

Lemma 4.3. Let X be a prescheme, let X' be the strict localization of X at a geometric point $\bar{x}$, and let $f : X' \rightarrow X$ be the canonical map. For $F \in \text{Ob } D_c^-(X)$ and $G \in \text{Ob } D^+(X)$, the canonical functorial homomorphism

$$f^* R \underline{\text{Hom}}(F, G) \longrightarrow R \underline{\text{Hom}}(f^* F, f^* G)$$

is an isomorphism.

Proof. As usual, one reduces to the case where F and G are concentrated in degree 0. The question being local near x , suppose X affine. The sheaf F admits a left resolution (SGA A, Exposé IX, 2.7) by sheaves of the form $i_!(A_U)$, where $i : U \rightarrow X$ is étale of finite presentation.

By dévissage, we are reduced to $F = i_!(A_U)$. Form the cartesian square

$$\begin{array}{ccc} U' & \xrightarrow{g} & U \\ \downarrow i' & & \downarrow i \\ X' & \xrightarrow{f} & X. \end{array}$$

There are canonical isomorphisms, trivial special cases of duality (SGA A, Exposé XVIII, 3.1.10),

$$\mathrm{R}\underline{\mathrm{Hom}}(F, G) \xleftarrow{\sim} \mathrm{R}i_*i^*G, \quad \mathrm{R}\underline{\mathrm{Hom}}(f^*F, f^*G) \xleftarrow{\sim} \mathrm{R}i'_*g^*i^*G.$$

Under these isomorphisms the canonical map is identified with the base-change map

$$f^*\mathrm{R}i_*i^*G \longrightarrow \mathrm{R}i'_*g^*i^*G,$$

and this is an isomorphism, as is seen immediately by passage to the limit (SGA A, Exposé VII, 5.11). $\square$

Lemma 4.4. Let X be a prescheme, and let $K \in \mathrm{Ob}\, \mathrm{D}_c^+(X)$ be a complex satisfying conditions (i) and (ii) of 1.7. Then K is dualizing if and only if, for every strict localization

$$f : \overline{X}(\overline{x}) \longrightarrow X,$$

the complex f^*K is dualizing.

Proof. Use the fact that every constructible sheaf on a strict localization $\overline{X}(\overline{x})$ is induced by a constructible sheaf on an étale $\overline{x}$ -pointed scheme over X (SGA A, Exposé IX, 2.7.4), and apply 4.3. $\square$

4.5

Let X be a prescheme, let $\overline{x}$ be a geometric point of X , and let

$$f : X' = \overline{X}(\overline{x}) \longrightarrow X$$

be the corresponding strict localization. Let Y be the integral closed sub-prescheme which is the closure of x in X ; write

$$g : \overline{x} \longrightarrow Y, \quad i : Y \longrightarrow X$$

for the canonical maps, and form the cartesian square

$$\begin{array}{ccc} \bar{x} & \xrightarrow{i'} & X' \\ g\downarrow & & \downarrow f \\ Y & \xrightarrow{i} & X. \end{array}$$

The point $\bar{x}$ is both the closed point of X' and the strict localization of Y at $\bar{x}$.

We denote by

$$\mathrm{R}\Gamma_{\bar{x}} : \mathrm{D}^+(X) \longrightarrow \mathrm{D}^+(\bar{x})$$

the functor defined by

$$\mathrm{R}\Gamma_{\bar{x}} = g^* \mathrm{R}^! i. \quad (4.5.1)$$

Using the well-known relation between the functors $\mathrm{R}^! i$ and $\mathrm{R} j_* j^*$, where $j : U \rightarrow X$ is the open complement of Y , and applying passage to the limit (SGA A, Exposé VII, 5.11), one obtains a canonical isomorphism

$$\mathrm{R}\Gamma_{\bar{x}} \xrightarrow{\sim} \mathrm{R}^! i' f^*. \quad (4.5.2)$$

When x is a closed point with separably closed residue field, the functor $\mathrm{R}\Gamma_{\bar{x}}$ coincides with the usual functor $\mathrm{R}\Gamma_x$ (SGA A, Exposé V, 4.3), which justifies the notation.

Let $K_X \in \mathrm{Ob} \mathrm{D}^+(X)$, and put

$$K_Y = \mathrm{R}^! i(K_X), \quad K_{X'} = f^* K_X, \quad K_{\bar{x}} = \mathrm{R}\Gamma_{\bar{x}}(K_X).$$

Here $K_{\bar{x}}$ is not the fibre of K_X at $\bar{x}$. Suppose K_X dualizing. Then by 1.15 and 4.4, the complexes $K_{X'}$, K_Y , and $K_{\bar{x}}$ are dualizing. Hence by 2.1,

$$K_{\bar{x}} \simeq A[d]$$

for some $d \in \mathbb{Z}$. We say that K_X is *normalized at $\bar{x}$* if $d = 0$ and an isomorphism $K_{\bar{x}} \simeq A$ has been chosen. When x is closed with separably closed residue field, this recovers the notion introduced in 4.1.

Let K_X be dualizing and put

$$\underline{D}_X = \mathrm{R} \underline{\mathrm{Hom}}(, K_X), \quad \underline{D}_Y = \mathrm{R} \underline{\mathrm{Hom}}(, K_Y),$$

etc. For $F \in \mathrm{Ob} \mathrm{D}_c^b(X)$, let us compute $(\underline{D}_X F)_{\bar{x}}$. We have

$$\begin{aligned} (\underline{D}_X F)_{\bar{x}} &\xrightarrow{\sim} g^* i^* \underline{D}_X F \\ &\xrightarrow{\sim} g^* \underline{D}_Y \mathrm{R}^! i F && \text{by the complementary induction formula (4.2.1)} \\ &\xrightarrow{\sim} \underline{D}_{\bar{x}} g^* \mathrm{R}^! i F && \text{by 4.3,} \end{aligned}$$

that is,

$$(\underline{D}_X F)_{\bar{x}} \xrightarrow{\sim} \underline{D}_{\bar{x}} \mathrm{R}\Gamma_{\bar{x}}(F). \quad (4.5.3)$$

Of course, one could make the same calculation through X' rather than through Y , and would obtain the canonical isomorphism

$$(\underline{D}_X F)_{\bar{x}} \xrightarrow{\sim} \underline{D}_{\bar{x}} \mathrm{R}^! i' f^* F,$$

compatible with (4.5.3) under (4.5.2).

If, moreover, K_X is normalized at $\bar{x}$, one obtains perfect pairings generalizing (4.2.2') and (4.2.2''):

$$\underline{H}_Y^i(F)_{\bar{x}} \times R^{-i} \Gamma_{\bar{x}}(F) \longrightarrow A, \quad (4.5.3')$$

$$\underline{H}^i((\underline{D}_X F)_{\bar{x}}) \times \underline{H}_{\bar{x}}^{-i}(F) \longrightarrow A, \quad (4.5.3'')$$

where $\underline{H}_{\bar{x}}^i = \underline{H}^i \mathrm{R}\Gamma_{\bar{x}}$.

Conversely, let $K_X \in \mathrm{Ob} \, \mathrm{D}^+(X)$ satisfy conditions (i) and (ii) of 1.7, and suppose that

$$K_{\bar{x}} = \mathrm{R}\Gamma_{\bar{x}}(K_X)$$

is dualizing for every geometric point $\bar{x}$ of X . Then, for each $\bar{x}$, one can define a canonical map, cf. 1.12 b)(ii),

$$\underline{D}_X(F)_{\bar{x}} \longrightarrow \underline{D}_{\bar{x}} \mathrm{R}\Gamma_{\bar{x}}(F), \quad F \in \mathrm{Ob} \, \mathrm{D}_c^b(X). \quad (4.5.3 \text{ bis})$$

If this map is an isomorphism for all $\bar{x}$ and all F , then K_X is dualizing. Indeed, one must verify that

$$F \longrightarrow \underline{D}_X \underline{D}_X(F)$$

is an isomorphism for all $F \in \mathrm{Ob} \, \mathrm{D}_c^b(X)$. At an arbitrary geometric point $\bar{x}$, one has

$$\begin{aligned} F_{\bar{x}} &\xrightarrow{\sim} \underline{D}_{\bar{x}} \underline{D}_{\bar{x}}(F_{\bar{x}}) && \text{since } K_{\bar{x}} \text{ is dualizing} \\ &\xrightarrow{\sim} \underline{D}_{\bar{x}} \mathrm{R}\Gamma_{\bar{x}} \underline{D}_X(F) && \text{by 4.3 and ordinary induction} \\ &\xrightarrow{\sim} \underline{D}_{\bar{x}} \underline{D}_{\bar{x}}((F)_{\bar{x}}) && \text{by hypothesis.} \end{aligned}$$

Modulo the verification of compatibility, this proves that (F, K_X) is bidualizing.

Proposition 4.5.4. Let X be a prescheme, and let $K_X \in \mathrm{Ob} \, \mathrm{D}^+(X)$ satisfy conditions (i) and (ii) of 1.7. Then K_X is dualizing if and only if, for every geometric point $\bar{x}$ of X , the complex $K_{\bar{x}}$ is dualizing and the canonical map (4.5.3 bis) is an isomorphism for every $F \in \mathrm{Ob} \, \mathrm{D}_c^b(X)$.

Remark 4.5.5. The existence of dualizing complexes must be regarded, in a sense, as an important complement to the global duality theorem. The latter says that, for a separated

morphism of finite type $f : X \rightarrow Y$ with Y quasi-compact, there is a canonical isomorphism

$$\underline{D}_Y R_! f(F) \xrightarrow{\sim} R_! f_* \underline{D}_X(F)$$

with the notation of 1.12. To use this in the calculation of $R_! f(F)$, when $F \in \text{Ob } D_c^b(X)$, one needs information about $\underline{D}_X(F)$. If K_X is dualizing, this information is supplied by (4.5.3), which allows, at least theoretically, the fibres of $\underline{D}_X(F)$ to be computed as cohomological invariants of a purely spatial nature.

Exercise 4.5.6. Let X be a prescheme endowed with a dualizing complex (1.7). The family of functors

$$\underline{H}_{\bar{x}}^i : D_c^b(X) \longrightarrow A\text{-modules},$$

where i runs through $\mathbb{Z}$ and $\bar{x}$ runs through the geometric points of X , is conservative: a morphism $F \rightarrow G$ in $D_c^b(X)$ such that all maps

$$\underline{H}_{\bar{x}}^i(F) \longrightarrow \underline{H}_{\bar{x}}^i(G)$$

are isomorphisms is itself an isomorphism.

4.6

Let X be a prescheme, let $j : Y \rightarrow X$ be a closed sub-prescheme, and let $i : \bar{x} \rightarrow Y$ be a geometric point of Y . Let K_X be a dualizing complex on X , normalized at $\bar{x}$ in the sense of 4.5. We have seen in 4.5 that, when Y is the closure of x , for $F \in \text{Ob } D_c^b(X)$ there is a perfect pairing

$$R^! j(F)_{\bar{x}} \times \underline{D}_X(F)_{\bar{x}} \longrightarrow A.$$

It is not difficult to define an analogous pairing in the general case. Factor i as

$$\bar{x} \xrightarrow{i'} \{\bar{x}\} \xrightarrow{i''} Y.$$

For $F \in \text{Ob } D_c^b(X)$, one has canonical isomorphisms

$$\begin{aligned} R\Gamma_{\bar{x}} j^* \underline{D}_X(F) &\xrightarrow{\sim} R\Gamma_{\bar{x}} \underline{D}_Y R^! j(F) \\ &\xrightarrow{\sim} i'^* R^! i''^* \underline{D}_Y R^! j(F) \\ &\xrightarrow{\sim} \underline{D}_{\bar{x}} i'^* R^! j(F), \end{aligned}$$

using the complementary induction formula, the definition of $R\Gamma_{\bar{x}}$, ordinary induction, and 4.3. Thus, finally, one obtains a canonical isomorphism

$$R^! j(F)_{\bar{x}} \xrightarrow{\sim} \underline{D}_{\bar{x}} R\Gamma_{\bar{x}} j^* \underline{D}_X(F), \quad (4.6.1)$$

and hence perfect pairings

$$\begin{aligned} \mathrm{R}^!j(F)_{\bar{x}} \times \mathrm{R}\Gamma_{\bar{x}} j^* \underline{D}_X(F) &\longrightarrow A, \\ \underline{H}_Y^i(F)_{\bar{x}} \times \underline{H}_{\bar{x}}^{-i}((\underline{D}_X F)|Y) &\longrightarrow A. \end{aligned} \quad (4.6.1'')$$

Example 4.6.2. Let X be a regular prescheme, let $\bar{x}$ be a geometric point of X , and put $d = \dim \mathcal{O}_{X, \bar{x}}$. Let

$$K_X = (\mu_n^{\otimes d})_X[2d].$$

If X satisfies (reg) (3.1.1) and the purity theorem (3.1.4), then K_X is canonically normalized at $\bar{x}$. Hence, if K_X is dualizing, one has perfect pairings

$$\underline{H}_Y^i(F)_{\bar{x}} \times \underline{H}_{\bar{x}}^{2d-i}(F'|Y) \longrightarrow A,$$

for every closed subset Y containing $\bar{x}$ and every $F \in \mathrm{Ob} \, \mathrm{D}_c^b(X)$, where

$$F' = \mathrm{R} \underline{\mathrm{Hom}}\left(F, (\mu_n^{\otimes d})_X\right).$$

We end with a strictly local variant of (4.6.1).

4.7

Let X be a noetherian strictly local scheme (SGA A, Exposé VIII, 4.2), with closed point x . Let Y be a non-empty closed subset of X , and consider the canonical immersions

$$U = X - Y \xrightarrow{i} V = X - \{x\} \xrightarrow{j} X.$$

Let K_X be a dualizing complex normalized at x (4.1), so that there are corresponding dualizing complexes on U, V, Y, x ; by hypothesis

$$\underline{D}_x = \mathrm{R} \mathrm{Hom}(\ , A),$$

and

$$K_V = j^! K_X, K_U = i^* K_V, \quad K_Y = \mathrm{R}\Gamma_Y(K_X).$$

Let $G \in \mathrm{Ob} \, \mathrm{D}_c^b(U)$, and put $G' = \underline{D}_U(G)$. We shall define a natural pairing

$$\mathrm{R}\Gamma(U, G) \times \mathrm{R}\Gamma(V, i_* G')[-1] \longrightarrow A, \quad (4.7.1)$$

or equivalently a functorial homomorphism

$$\mathrm{R}\Gamma(U, G) \overset{L}{\otimes}_A \mathrm{R}\Gamma(V, i_* G')[-1] \longrightarrow A. \quad (4.7.1')$$

The ring $A = \mathbb{Z}/n\mathbb{Z}$ has infinite cohomological dimension for $\ell \geq 2$, so at this point one only knows how to define derived tensor products

$$\overset{L}{\otimes} : D^-(A) \times D^-(A) \rightarrow D^-(A)$$

and

$$\overset{L}{\otimes} : D(A) \times D^b(A)_{\text{torf}} \rightarrow D(A).$$

We leave to the reader the task of imposing convenient hypotheses ensuring that the two factors in (4.7.1') belong to $D^b(A)$ or that one of them has finite Tor-dimension.⁴

First note that, by the complementary duality formula (1.12 a)(ii), there is a canonical isomorphism

$$i_!(G') \xrightarrow{\sim} \underline{D}_V(Ri_*G). \quad (4.7.2)$$

On the other hand, there is the ordinary duality isomorphism

$$\text{R Hom}(A_V, G) \xrightarrow{\sim} \text{R Hom}(A_V, Ri_*G). \quad (4.7.3)$$

By the definition of $\underline{D}_V$, and independently of the fact that K_V is dualizing, there is a canonical map

$$Ri_*G \overset{L}{\otimes} \underline{D}_V(Ri_*G) \longrightarrow K_V. \quad (4.7.4)$$

From (4.7.2), (4.7.3), and (4.7.4) one obtains a canonical map

$$\text{R}\Gamma(U, G) \overset{L}{\otimes}_A \text{R}\Gamma(V, i_*G') \longrightarrow \text{R Hom}(A_V, K_V). \quad (4.7.5)$$

But $K_V = j^!K_X$, and there is the duality isomorphism (SGA A, 3.1.10)

$$\text{R Hom}(A_V, K_V) \xrightarrow{\sim} \text{R Hom}(A_{V,X}, K_X). \quad (4.7.6)$$

Finally the exact sequence

$$0 \longrightarrow A_{V,X} \longrightarrow A_X \longrightarrow A_x \longrightarrow 0 \quad (4.7.7)$$

gives a homomorphism of degree +1,

$$\text{R Hom}(A_{V,X}, K_X) \longrightarrow \text{R Hom}(A_x, K_X) \xrightarrow{\sim} A. \quad (4.7.8)$$

Composing (4.7.5), (4.7.6), and (4.7.8) gives the desired map (4.7.1').

We now see that the pairing (4.7.1) is perfect, i.e. that (4.7.1') identifies $\text{R}\Gamma(V, i_*G')[-1]$

⁴Added in 1977: this assertion is inaccurate; in fact it is easy to extend $\overset{L}{\otimes} : D^-(A) \times D^-(A) \rightarrow D^-(A)$ to $\overset{L}{\otimes} : D^-(A) \times D(A) \rightarrow D(A)$.

with $\underline{D}_x \mathrm{R}\Gamma(U, G)$. Suppose, as we may, that

$$G = (ji)^* F$$

for some $F \in \mathrm{Ob} \, \mathrm{D}_c^b(X)$. Put $F' = \underline{D}_X(F)$ and $F'_{U,X} = (ji)_! G'$. There is an exact sequence

$$0 \longrightarrow F'_{U,X} \longrightarrow F' \longrightarrow F'|Y \longrightarrow 0, \quad (4.7.9)$$

giving a distinguished triangle

$$\mathrm{R}\Gamma_x(F'_{U,X}) \rightarrow \mathrm{R}\Gamma_x(F') \rightarrow \mathrm{R}\Gamma_x(F'|Y) \xrightarrow{[1]}.$$

By the definition of $\mathrm{R}\Gamma_x$,

$$\mathrm{R}\Gamma_x(F'_{U,X}) = \mathrm{R}\mathrm{Hom}(A_x, F'_{U,X}).$$

Using (4.7.7) and the fact that $(F'_{U,X})_x = 0$, one gets a canonical isomorphism

$$\mathrm{R}\mathrm{Hom}(A_{V,X}, F'_{U,X})[1] \xrightarrow{\sim} \mathrm{R}\Gamma_x(F'_{U,X}).$$

But

$$\begin{aligned} \mathrm{R}\mathrm{Hom}(A_{V,X}, F'_{U,X}) &= \mathrm{R}\mathrm{Hom}(j_! A_V, j_! i_! G') \\ &\xrightarrow{\sim} \mathrm{R}\mathrm{Hom}(A_V, i_! G') \quad \text{by duality (SGA A, 3.1.10),} \end{aligned}$$

and hence there is a canonical isomorphism

$$\mathrm{R}\Gamma(V, i_! G')[-1] \xrightarrow{\sim} \mathrm{R}\Gamma_x(F'_{U,X}). \quad (4.7.10)$$

On the other hand, the exact sequence

$$0 \longrightarrow A_{U,X} \longrightarrow A_X \longrightarrow A_Y \longrightarrow 0 \quad (4.7.11)$$

gives a distinguished triangle

$$\mathrm{R}\Gamma_Y(F) \rightarrow \mathrm{R}\Gamma(X, F) \rightarrow \mathrm{R}\Gamma(U, F|U) \xrightarrow{[1]}.$$

By (4.7.1) and (4.7.10), one has a natural pairing

$$\mathrm{R}\Gamma(U, F|U) \times \mathrm{R}\Gamma_x(F'_{U,X}) \longrightarrow A. \quad (4.7.12)$$

By (4.6.1), one has a perfect pairing

$$\mathrm{R}\Gamma_Y(F) \times \mathrm{R}\Gamma_x(F'|Y) \longrightarrow A. \quad (4.7.13)$$

Finally, by ordinary induction (1.12 b)(i)), one has a perfect pairing

$$\mathrm{R}\Gamma(X, F) \times \mathrm{R}\Gamma_x(F') \longrightarrow A. \quad (4.7.14)$$

The reader patient enough to orient himself through this maze of “canonical arrows” will verify that the pairings (4.7.12), (4.7.13), and (4.7.14) are compatible with the arrows of the two triangles, and hence that (4.7.12), and therefore (4.7.1), are perfect.

The pairing of the two triangles gives two exact sequences transposed to one another:

$$\cdots \rightarrow \underline{H}_Y^i(X, F) \rightarrow \underline{H}^i(X, F) \rightarrow \underline{H}^i(U, F) \rightarrow \cdots$$

and

$$\cdots \leftarrow \underline{H}_x^{-i}(Y, F'|Y) \leftarrow \underline{H}_x^{-i}(X, F') \leftarrow \underline{H}_x^{-i-1}(V, F'_{U,V}) \leftarrow \cdots .$$

Proposition 4.7.16. Let X be a noetherian strictly local scheme, with closed point x . Let Y be a non-empty closed subset of X , put $U = X - Y$ and $V = X - x$, and let $i : U \rightarrow V$ and $j : V \rightarrow X$ be the canonical inclusions. Let K_X be a dualizing complex on X normalized at x , hence giving associated dualizing complexes on V , U , and Y .

For $G \in \text{Ob } D_c^b(U)$, there is a perfect pairing

$$\text{R}\Gamma(U, G) \times \text{R}\Gamma(V, i_* G')[-1] \rightarrow A,$$

where $G' = \underline{D}_U(G)$. If $G = F|U$ for $F \in \text{Ob } D_c^b(X)$, then there is a canonical isomorphism

$$\text{R}\Gamma(V, i_* G')[-1] \xrightarrow{\sim} \text{R}\Gamma_x(F'_{U,X}),$$

where $F' = \underline{D}_X(F)$ and $F'_{U,X} = (ji)_!(F'|U) = (ji)_!(G')$, and the exact localization triangles are paired perfectly as described above.

Remark 4.7.17. If, in the preceding proposition, X is a strictly local regular prescheme of dimension d , if $Y = \{x\}$, and if $K_X = (\mu_n^{\otimes d})_X[2d]$, then $K_U = (\mu_n^{\otimes d})_U[2d]$. Assuming K_X dualizing, local duality on X implies, for every locally constant constructible A -module G , perfect pairings

$$H^i(U, G) \times H^{2d-1-i}(U, G^\circ) \longrightarrow A,$$

where $G^\circ = \underline{\text{Hom}}(G, (\mu_n^{\otimes d})_U)$. Formally, this corresponds to Poincaré duality on a scheme of dimension $2d - 1$.

5. Local duality on curves

Theorem 5.1. Let X be a noetherian regular prescheme of dimension 1. Suppose n is prime to the residual characteristics of X . Then the purity theorem is true on X (3.1.4), and the complex A_X is dualizing (1.7).

Proof. Assume first that purity has been proved. Since X obviously satisfies (reg), and since

(cdloc) holds by SGA A, Exposé X, 2.2, 3.2.1 gives

$$\dim . q. inj(A_X) = 2.$$

Moreover condition (ii) of 1.7 holds by 3.3.1 b)(i), whose proof uses purity. Hence, by 4.4, we are reduced to proving that A_X is dualizing when X is strictly local. For purity, also a local question near a closed point, we may likewise suppose X strictly local. Thus it remains to prove the theorem for strictly local X .

Assume then that X is strictly local. Let x be its closed point, let p be the residual characteristic, let η be its generic point, and put

$$U = X - x = \text{Spec}(k(\eta)).$$

Let $G = \text{Gal}(\overline{k(\eta)}/k(\eta))$ be the fundamental group of U . One knows (cf. Chapter IV, §2, Exercise 2) that there is a canonical exact sequence

$$1 \longrightarrow P \longrightarrow G \longrightarrow \mathbb{Z}_p(1) \longrightarrow 1, \quad (5.1.1)$$

where

$$\mathbb{Z}_p(1) = \varprojlim \mu_{p^r}(\overline{k(\eta)})$$

and P is a pro- p -group. Of course, the quotient is isomorphic, but not canonically, to the product of the $\mathbb{Z}_\ell(1)$ for $\ell \neq p$.

On the other hand, the étale cohomology of U is interpreted as the Galois cohomology of G (SGA A, Exposé VIII, 2); sheaves, respectively constructible sheaves, of A -modules correspond to A -modules, respectively finite-type A -modules, on which G acts continuously.

The groups $H_X^i(A_X)$ are determined by the exact sequence

$$0 \rightarrow H_X^0(A_X) \rightarrow H^0(A_X) \rightarrow H^0(U, A_U) \rightarrow H_X^1(A_X) \rightarrow 0, \quad (5.1.2)$$

and by the isomorphisms

$$H_X^i(A_X) \xrightarrow{\sim} H^{i-1}(U, A) \quad (i \geq 2). \quad (5.1.3)$$

Trivially, $H^0(X, A_X) = H^0(U, A_U) = A$, so $H_X^0(A_X) = H_X^1(A_X) = 0$.

For a Galois G - A -module M , the Hochschild–Serre spectral sequence gives

$$E_2^{pq} = H^p(H, H^q(P, M)) \Rightarrow H^{p+q}(G, M).$$

Since n is prime to p and P is a pro- p -group, $H^q(P, M) = 0$ for $q \neq 0$, hence

$$H^i(G, M) \xrightarrow{\sim} H^i(H, M^P). \quad (5.1.4)$$

An analogous argument identifies $H^i(H, N)$ with $H^i(\mathbb{Z}_p(1), N)$ when the complementary factors act trivially on N . Thus, with the preceding convention, there is a canonical isomorphism

$$H^i(G, M) \xrightarrow{\sim} H^i(\mathbb{Z}_p(1), M^P). \quad (5.1.5)$$

The Galois cohomology of $\mathbb{Z}_p(1)$ is well known (CG, Chapter I, p. 31): for a $\mathbb{Z}_p(1)$ - A -module N ,

$$H^i(\mathbb{Z}_p(1), N) = \begin{cases} N^{\mathbb{Z}_p(1)}, & i = 0, \\ N_{\mathbb{Z}_p(1)}, & i = 1, \\ 0, & i \geq 2. \end{cases} \quad (5.1.6)$$

It follows from (5.1.3), (5.1.5), and (5.1.6) that

$$H^i(U, A_X) = 0 \quad (i > 2),$$

and $H^2(U, A_X) \simeq A$. This last isomorphism is not canonical, since it depends on a choice of an identification of $\mathbb{Z}(1)$ with $\mathbb{Z}/n\mathbb{Z}$. But using the Kummer exact sequence, one obtains (SGA A, Exposé XIX, 1.3) a canonical isomorphism, given by the local fundamental class,

$$H^2(U, \mu_n) \xrightarrow{\sim} A. \quad (5.1.7)$$

Purity is therefore proved.

It remains to show that A_X is dualizing. By 4.7.17, it suffices to show that, for every locally constant constructible A_X -module M , the pairing

$$\mathrm{R}\Gamma(U, M) \times \mathrm{R}\Gamma(U, \underline{\mathrm{Hom}}(M, A_X)) \longrightarrow \mathrm{R}\Gamma(U, A_X) \xrightarrow{\sim} A[-2] \quad (5.1.8)$$

is a perfect duality. We interpret this pairing in terms of Galois cohomology, using the following elementary lemma.

Lemma 5.1.9. Let $D = \mathrm{Hom}(_, A)$ be the Pontryagin-duality functor on the category of finite-type A -modules. Let M be a finite-type A -module on which a group G acts A -linearly.

- (i) There is a canonical isomorphism $M^G \xrightarrow{\sim} D(M_G)$.
- (ii) If G is a profinite group acting continuously on M and if G has pro-order prime to n , then the canonical map $M^G \rightarrow M_G$ is an isomorphism.

Proof. By definition,

$$M^G = \varprojlim_{g \in G} \mathrm{Ker}(M \xrightarrow{g-1} M),$$

hence

$$D(M^G) \xrightarrow{\sim} \varprojlim_{g \in G} \mathrm{Coker}(DM \xrightarrow{g-1} DM) \xrightarrow{\sim} (DM)_G,$$

which proves (i).

For (ii), let M' be defined by

$$0 \longrightarrow M' \longrightarrow M \longrightarrow M_G \longrightarrow 0.$$

Since G has pro-order prime to n , one has $H^0(G, M') = 0$. The cohomology exact sequence implies that $M^G \rightarrow M_G$ is an epimorphism. Applying D and using (i), one sees that it is also a monomorphism. $\square$

Applying 5.1.9 to M and to $D(M)$, one obtains a canonical identification

$$\mathrm{Hom}(M, A)^G \simeq \mathrm{Hom}(M^G, A).$$

After choosing an isomorphism $\mathbb{Z}(1) \simeq \mathbb{Z}/n\mathbb{Z}$, the pairing (5.1.8) can be written, using (5.1.5), as

$$H^i(G, M) \times H^{1-i}(G, D(M^P)) \longrightarrow H^1(\mathbb{Z}_p(1), A) \simeq A. \quad (5.1.8')$$

It follows immediately from (5.1.6) and 5.1.9(i) that this is a perfect pairing. This completes the proof of 5.1.

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Appendix to Exposé I

by *L. Illusie*

The preceding theory extends without difficulty to base rings more general than the constant ring $\mathbb{Z}/n\mathbb{Z}$. We briefly indicate how the theory is written for locally constant base rings, annihilated by an integer $n > 0$, whose geometric fibres are noetherian rings. This is only a provisional framework: sooner or later one will have to globalize, taking as “coefficients” on a prescheme X relative preschemes on the site $(X_{\text{ét}}, \mathbb{Z}/n\mathbb{Z})$.

1. Constructible sheaves

1.1. Let X be a prescheme, and let A_X be a sheaf of rings on X for the étale topology, locally constant, with geometric fibres noetherian on the left. Write $\underline{\text{Mod}}_{A_X}$ for the category of left A_X -modules, and $D(X)$, or $D(A_X, X)$ if ambiguity is possible, for the corresponding derived category. The notion of a constructible left A_X -module is defined as in SGA A, Exposé IX, 2.3, replacing the expression “locally constant with value of finite presentation” by “of finite presentation as an A_X -module.”

Let $D_c(X)$ be the full subcategory of $D(X)$ consisting of complexes E such that $H^i(E)$ is a constructible A_X -module for every i . It is a triangulated subcategory of $D(X)$.

1.2. Let $E \in \text{Ob } D(X)$. One says, cf. Exposé II, 3.11, that E is pseudo-coherent if, for every i , the sheaf $H^i(E)$ is an A_X -module of finite presentation, or equivalently locally constant constructible. We denote by $D_{\text{coh}}(X)$, or $D_{\text{coh}}(A_X, X)$, the full subcategory of $D(X)$ consisting of pseudo-coherent objects. It follows from SGA A, Exposé IX, 2.1 that this is a triangulated subcategory of $D_c(X)$.

Proposition 1.3, dévissage lemma. Let X be a quasi-compact and quasi-separated, respectively noetherian, prescheme, and let $\mathcal{C}$ be a strictly full subcategory of the category $\text{Cons}(X)$ of constructible left A_X -modules. Suppose $\mathcal{C}$ is stable under extensions. The following conditions are equivalent:

- (i) $\mathcal{C} = \text{Cons}(X)$;
- (ii) $\mathcal{C}$ contains sheaves of the form i_*F , where $i : Y \rightarrow X$ is a sub-prescheme of finite presentation, respectively an integral sub-prescheme, and F is locally constant constructible on Y .

If A_X is constant with value A , a left noetherian ring annihilated by $n > 0$, these conditions are equivalent to:

- (iii) $\mathcal{C}$ is stable under direct factors and contains sheaves of the form $f_*j_!(M_Y)$, where $j : U \rightarrow Y$ is an open immersion, respectively an open subset of an integral prescheme, $f : Y \rightarrow X$ is finite, respectively finite with the generic point of Y separable over its image, and M_Y is the constant left A_Y -module associated with a cyclic A -module annihilated by a prime divisor of n .

Finally, if X is noetherian and satisfies (reg) (I, 3.1.1), condition (ii), respectively (iii), is equivalent to (i) if Y , respectively U , is moreover assumed regular.

Proof. The equivalence of (i) and (ii) is proved as in the case where A_X is constant; see SGA A, Exposé IX, 2.4 and 2.5. Suppose A_X constant with value A , annihilated by n , and prove that (iii) implies (ii). Let $E \in \text{Ob Cons}(X)$; we show that $E \in \text{Ob } \mathcal{C}$. By the equivalence (i) $\Leftrightarrow$ (ii), one may suppose $E = j_! F$, where $j : Z \rightarrow X$ is the immersion of a sub-prescheme of finite presentation, respectively an integral sub-prescheme, and F is a locally constant constructible A_Z -module. After a further dévissage, we may suppose F trivialized by an étale covering $g : Z' \rightarrow Z$.

Then one has an exact sequence

$$0 \rightarrow F \rightarrow g_* g^* F \rightarrow \text{coker} \rightarrow 0,$$

and one is reduced to proving that $g_* g^* F$ and the cokernel belong to $\text{Ob } \mathcal{C}$. This follows at once from (iii) and the induction hypothesis on the dimension of Z , taking into account that $g_* g^* F$ is locally constant and that the cokernel is supported on the non-smooth locus of g . $\square$

Corollary 1.4, dévissage for pseudo-coherent objects. Let X be quasi-compact and quasi-separated, respectively noetherian, and let $\mathcal{C}$ be a strictly full triangulated subcategory of $D_{\text{coh}}(X)$. The following conditions are equivalent:

- (i) $\mathcal{C} = D_{\text{coh}}(X)$;
- (ii) $\mathcal{C}$ contains objects of the form $i_* F$, where $i : Y \rightarrow X$ is a sub-prescheme of finite presentation, respectively an integral sub-prescheme, and $F \in \text{Ob } D_{\text{coh}}(Y)$.

If A_X is constant with value A annihilated by n , these are equivalent to the condition that $\mathcal{C}$ is stable under direct factors and contains the objects $f_* j_!(M_Y)$ with j, f , and M_Y as in 1.3(iii).

1.5. With the notation of 1.1, a complex $K \in \text{Ob } D(X)$ is called *dualizing* if it satisfies conditions (i), (ii), (iii) of I, 1.7. Explicitly:

- (i) $K \in \text{Ob } D_c^b(X)$;
- (ii) for every $F \in \text{Ob } D_c^b(X)$, $\text{R } \underline{\text{Hom}}(F, K) \in \text{Ob } D_c^b(X)$;
- (iii) the canonical map

$$F \longrightarrow \text{R } \underline{\text{Hom}}(\text{R } \underline{\text{Hom}}(F, K), K)$$

is an isomorphism for $F \in \text{Ob } D_c^b(X)$.

Corollary 1.6, dévissage for dualizing complexes. Let X be a noetherian prescheme. Let $K \in \text{Ob } D_c^b(X)$ satisfy conditions (i) and (ii) of 1.5. Then K is dualizing if and only if, for every geometric point $\bar{x}$ of X , the complex $\text{R}\Gamma_{\bar{x}}(K)$ is dualizing.

Proof. This follows immediately from 1.4, dévissage for biduality, and from I, 4.3. □

1.7. Keep the preceding notation, but suppose A_X commutative. The notions of bidualizing pair and dualizing complex are defined as in I, 1.7. The statements I, 1.9, 1.11, and 1.14 remain valid without modification. The statements I, 1.12, 1.13, and 1.15 are also valid, provided A is annihilated by an integer $n > 0$ prime to the residual characteristics of X . Finally, the uniqueness theorem I, 2.1 is valid provided one assumes that $\text{Spec}(A_{\bar{x}})$ is connected for every geometric point $\bar{x}$ of X .

2. Dualizing complexes: definition and formal properties

Retaining the preceding notation but assuming A_X commutative, the definitions and formal statements of the preceding paragraph remain in force. In particular, the formalism of dualizing complexes is unchanged, subject to the same restriction that A be killed by an integer $n > 0$ invertible on the base and to the connectedness hypothesis on $\text{Spec}(A_{\bar{x}})$ for uniqueness.

3. Absolute purity and quasi-injective dimension: source fragment

The source packet for this appendix contains a discontinuous fragment after the beginning of §3. It returns to the situation of absolute cohomological purity (I, 3.1.4): if X is regular locally noetherian, if $n > 0$ is prime to the residual characteristics of X , and if $i : Y \rightarrow X$ is a regular sub-prescheme of codimension $d > 0$ at every point, the assertion that (Y, X) satisfies absolute purity is the assertion that the local fundamental class homomorphism of SGA 4^{1/2}, Cycle 2.2, is an isomorphism.

The same fragment then records the obvious inequality

$$\dim \text{top}_x(K) \leq \dim . q . \text{inj}_x(K),$$

and says that examples where equality holds will be seen below. It also restates, for this more general coefficient sheaf A_X , the formal definitions of dualizing complexes and the hypotheses under which the statements of Exposé I carry over.

Proposition 5.1, Appendix: bound for quasi-injective dimension. Let X be a regular locally noetherian prescheme, equipped with a locally constant sheaf of rings A_X , annihilated by an integer $n > 0$ prime to the residual characteristics of X , with geometric

fibres left noetherian. Suppose that X satisfies (reg) (I, 3.1.1) and (cdloc) $_n$ (I, 3.1.3), and that the strong form of the purity theorem is true on X (4.4). These hypotheses are all satisfied if X is smooth over a perfect field, or if X is excellent of characteristic zero.

Let $F \in \text{Ob } D_c^b(A_X)$, with the sheaves $\underline{H}^i(F)$ locally constant. Let x be a point of X such that $F_{\bar{x}} \neq 0$ and

$$\dim .inj_{A_{\bar{x}}}(F_{\bar{x}}) < +\infty.$$

Then

$$\dim .q.inj_x(F) \leq 2 \dim \mathcal{O}_{X,x} + \dim .inj_{A_{\bar{x}}}(F_{\bar{x}}).$$

Beginning of proof in the source packet. One repeats, step by step, the proof of I, 3.2.1. Let the right-hand side be N ; the first aim is to prove

$$\underline{\text{Ext}}^i(E, F)_{\bar{x}} = 0 \quad (i > N)$$

for every constructible A_X -module E . Since the question is local, dévissage (1.3) reduces the verification to $E = i_! M$, where $i : Y \rightarrow X$ is an integral regular sub-prescheme, closed in an open $D(f)$, of codimension d at every point of $D(f)$, and where M is a locally constant constructible A_Y -module. The continuation of this proof is not present in the extracted $\square$

Exposé III. Lefschetz Formula

Lefschetz Formula

by A. Grothendieck, written up by L. Illusie

In this exposé we prove the Lefschetz formula in étale cohomology for correspondences between complexes of sheaves, in Verdier's formulation. In an earlier version, distributed by the IHES about ten years before, the formula was stated under resolution hypotheses, but the required compatibilities were not proved. Deligne's finiteness theorems (SGA 4^{1/2}, Finiteness) made it possible to dispense with these hypotheses, provided one works over schemes separated and of finite type over a field. As for the compatibilities, the writer has checked them, painfully.

Let us say at once that the Lefschetz formula considered here, like the trace formula for constant coefficients (SGA 4^{1/2}, Cycle) which it generalizes, is essentially tautological: it expresses certain compatibilities in the formalism of duality. Roughly speaking, it says that, under suitable conditions, the trace of an endomorphism of a hypercohomology complex $\text{RF}(X, L)$, where L is a complex of sheaves on X , may be computed as the "integral" of certain cohomology classes on $X \times X$. In practice, the endomorphisms considered arise from correspondences $C \subset X \times X$, and the classes recovered on the product are supported on $C \cap \Delta$, where Δ is the diagonal. These classes are local in nature near the fixed points, which

in principle makes their computation easier. Precise statements are given below, notably in 4.2 and 4.7. In fact, the computation of these local classes in terms of more understandable invariants, without which the formula would be of little use, has so far been carried out only in special situations; see 4.10, 4.11, and the following exposé, where one treats in particular the Frobenius correspondence on a curve.

1. Reminders and notation

1.1. Let S be a scheme and let $f : X \rightarrow S$ be a separated morphism of finite type. Suppose given on S a sheaf of rings A_S for the étale topology; put $A_X = f^* A_S$, and similarly $A_Y = g^* A_S$ for an S -scheme $g : Y \rightarrow S$. We suppose A_S annihilated by an integer $n > 0$ invertible on S .

1.2. We recall briefly the principal statements of the duality formalism for coefficients A_X (SGA A, or Exposé I in the case of constant coefficients). For a separated morphism of finite type $f : X \rightarrow Y$, there is a functor $R^!f$, compatible with composition, and an adjunction with $Rf_!$. It is characterized, in the situations considered below, by the duality isomorphisms

$$R \operatorname{Hom}_Y(Rf_!E, F) \simeq R \operatorname{Hom}_X(E, R^!fF),$$

together with their sheafified analogues.

1.3. Transitivity. For a composite $f = f_1 \circ f_2$, one has canonical transitivity isomorphisms

$$R^!f \simeq R^!f_2 R^!f_1,$$

compatible in the usual way for triple composites.

1.4. Duality for a separated finite-type morphism. For $f : X \rightarrow Y$ separated and of finite type, and for complexes of finite tor-dimension in the constructible derived category, one uses the usual duality isomorphisms relating $Rf_!$, Rf_* , and $R^!f$. These isomorphisms are the formal tools from which the Lefschetz construction below will be assembled.

1.5. Smooth case. If f is smooth, purely of relative dimension d , then there is a canonical isomorphism, the Poincaré duality formula,

$$R^!f(A_Y) \simeq A_X(d)[2d],$$

where $A_X(d)$ denotes the usual Tate twist. More generally this identifies the relative dualizing object in the smooth case.

1.6. External tensor products. For two separated finite-type morphisms $f_i : X_i \rightarrow S$ ($i = 1, 2$), there is an external tensor product on complexes and on cohomological correspondences. The compatibilities of this product with $Rf_!$, Rf_* , and duality are those needed for taking products of trace classes.

2. Lefschetz formula for a proper morphism

2.1. Let S be a scheme, and let X and Y be two separated finite-type S -schemes. With coefficients as in 1.1, let

$$E \in \text{Ob } D_{\text{ctf}}(X, A_X), \quad F \in \text{Ob } D_{\text{ctf}}(Y, A_Y).$$

One defines a canonical morphism, functorial in E and F , which is the basic cup-product pairing underlying the trace formalism.

2.2. For every S -morphism $u : E \rightarrow F$, lying over an S -morphism $g : X \rightarrow Y$, one defines the trace of u as an element of a group of the form

$$H^0\left(S, R \text{Hom}(Rf_{1!}E, Rf_{2*}F)\right),$$

where the f_i are the relevant structure morphisms. This description is intentionally functorial, since the formula will later be applied to morphisms carried by correspondences.

2.3. Suppose now $X = Y$, and let $g : X \rightarrow X$ be an S -endomorphism. For complexes $E, F \in \text{Ob } D_{\text{ctf}}(X, A_X)$ and a morphism

$$u : g^*E \longrightarrow F,$$

one obtains a trace element attached to u . Its local form is the class that appears in the Lefschetz formula.

2.4. Let, for $i = 1, 2$, $f_i : X_i \rightarrow Y_i$ be an S -morphism, with complexes $E_i, F_i \in \text{Ob } D_{\text{ctf}}^b(X_i)$ and $P_i, Q_i \in \text{Ob } D_{\text{ctf}}^b(Y_i)$. The formalism of external products gives the compatibility of traces with products. This is the categorical expression of the multiplicativity of Lefschetz classes.

2.5. The principal proposition of this section states the compatibility of the above trace construction with composition and exterior tensor product. In the special case $Y_1 = Y_2 = S$ and $X_1 = X_2 = X$, for an S -endomorphism g of X , it produces the Lefschetz class associated with a pair of mutually transposed cohomological correspondences.

2.6. The proof of 2.5 occupies the rest of section 2. It follows from the compatibility of external products with the duality isomorphisms that if $w_i : L_i \rightarrow M_i$ are morphisms of type 1 over f_i , then the exterior tensor product

$$w = w_1 \overset{L}{\otimes}_S w_2$$

makes the required diagram commute. The source text records the same verification twice, with notational variants u_i and w_i .

2.7–2.9. When $S = \operatorname{Spec}(k)$, with k separably closed, one has $H^0(S, A_S) = A$, and the Lefschetz formula takes its classical numerical form. In the case where E is a locally constant sheaf of free modules of finite rank and g is the identity, it reduces to the Euler–Poincaré formula: the trace is the alternating sum of traces on cohomology.

Exposé III continued. Cohomological correspondences and trace formulae

3. Cohomological correspondences

3.1. Generalities

We resume the notation of 2.2. Let $(S, \mathcal{O}_S)$ be a scheme, let X and Y be S -schemes, and let

$$L \in \operatorname{Ob} D^-(X), \quad M \in \operatorname{Ob} D^-(Y).$$

For $i \in \mathbb{Z}$ put

$$\operatorname{Hom}_S^i(L, M) = \operatorname{Ext}_{\mathcal{O}_S}^i(L, M), \tag{3.1.1}$$

where the $\operatorname{Ext}_{\mathcal{O}_S}^i$ are the derived functors of $\operatorname{Hom}_{\mathcal{O}_S}$. If $S = \operatorname{Spec}(k)$, with k a field, and if L and M are coherent sheaves, this is the ordinary group of extensions over S .

The Yoneda product gives pairings

$$\operatorname{Hom}_S^i(L, M) \otimes \operatorname{Hom}_S^j(M, N) \longrightarrow \operatorname{Hom}_S^{i+j}(L, N), \tag{3.1.2}$$

and hence, in particular, canonical homomorphisms

$$\operatorname{Hom}_S^i(L, M) \circ \operatorname{Hom}_{\mathcal{O}_S}(\operatorname{Ext}_S^j(M, N), \operatorname{Ext}_S^{i+j}(L, N)). \tag{3.1.3}$$

This is the convention by which the action of a correspondence on cohomology is read below.

3.2. Correspondences with support

Suppose given a commutative diagram

$$\begin{array}{ccc} C & \xrightarrow{p} & X \\ \downarrow q & & \downarrow f \\ Y & \xrightarrow{g} & S. \end{array}$$

For $L \in \text{Ob } D^-(X)$ and $M \in \text{Ob } D^-(Y)$ one has the formal morphism, as in 2.2.2,

$$p^*L \otimes_{\mathcal{O}_C}^L q^*M \longrightarrow p^*L \otimes_{\mathcal{O}_C}^L q^*M, \quad (3.2.1)$$

which, by adjunction, yields

$$\text{Hom}_C(p^*L, q^*M) \circ \text{Hom}_X(L, \text{Rp}_*q^*M). \quad (3.2.2)$$

Definition 3.3. A *cohomological correspondence* from L to M , with support in C , is an element

$$u \in \text{Hom}_C(p^*L, q^*M) = H^0\left(C, \underline{\text{RHom}}(p^*L, q^*M)\right).$$

3.3.1. Composition. Let $u \in \text{Hom}_C(p^*L, q^*M)$ and $v \in \text{Hom}_D(p^*M, q^*N)$ be correspondences for compatible supports. Their composite vu is obtained by pulling u to the intermediate correspondence, composing with v , and applying the relevant adjunction. This convention agrees with the Yoneda product (3.1.2) on the cohomology groups on which the correspondences act.

3.4. Base change. For a cartesian square of supports

$$\begin{array}{ccc} C' & \longrightarrow & C \\ \downarrow & & \downarrow \\ X' \times_S Y' & \longrightarrow & X \times_S Y, \end{array}$$

there is a canonical base-change morphism for cohomological correspondences. The construction is the evident one: pull back the two factors, use the functorial base-change morphism for the derived inverse images, and transport the result through adjunction.

Proposition 3.5. The base-change diagrams just described are commutative. Equivalently, the formation of the morphism induced by a cohomological correspondence is compatible with passage to a cartesian base change.

3.6. Proper smooth schemes over an algebraically closed field. Assume $S = \text{Spec}(k)$ with k algebraically closed. Let X and Y be proper smooth k -schemes, and let

$L \in \text{Ob } D_{\text{coh}}^b(X)$, $M \in \text{Ob } D_{\text{coh}}^b(Y)$. The duality isomorphisms give, in the notation of the source,

$$\text{Hom}(L, \mathcal{O}_X) \xrightarrow{\sim} H^*(X, L)^\vee. \quad (3.6.1)$$

Consequently a cohomological correspondence determines, by functoriality, a homomorphism on cohomology.

3.6.1. Cycle class case. If $L = M = \mathcal{O}_X$, $X = Y$, and $C \subset X \times X$ is a cycle of dimension $d = \dim X$, the associated cohomological correspondence is represented by the cohomology class

$$[C] \in H^{2d}(X \times X)(d),$$

and the operation induced on cohomology is cup-product with $[C]$, followed by the appropriate projection.

Direct image of a correspondence

3.7. Let

$$\begin{array}{ccccc} & C & & & \\ & \downarrow c & \searrow & & \\ X \times_S Y & \xrightarrow{p_1} & X & \xrightarrow{f} & S \\ & \xrightarrow{p_2} & Y & \xrightarrow{g} & S \end{array}$$

be the corresponding diagram. If

$$u \in \text{Hom}_C(p_1^* L, p_2^! M)$$

is a cohomological correspondence, its *direct image* is the composite

$$L \xrightarrow{\text{adj}} \text{R}f_* f^* L \xrightarrow{\text{R}f_*(u)} \text{R}f_* p_{2*} p_2^! M \xrightarrow{\text{trace}} M. \quad (3.7.1)$$

3.7.1. There is a canonical morphism

$$g^! f_* K \longrightarrow f'_* g'^! K, \quad (3.7.2)$$

which gives rise to the commutativity diagrams needed for direct images. In the situations used here it is the transpose of the usual base-change morphism.

3.8–3.9. One defines in the same way the direct-image correspondence under a proper morphism. If $X \xrightarrow{u} Y \xrightarrow{g} S$ is a composite, the resulting operations satisfy

$$(gf)_* = g_* f_*.$$

This is a direct verification from adjunction and the transitivity of the trace morphism.

Lefschetz formula

3.10. Let X , Y , and Z be S -schemes, and let

$$C \subset X \times_S Y, \quad D \subset Y \times_S Z$$

be correspondences. Their composite $D \circ C$ is defined by the usual fiber product over Y and projection to $X \times_S Z$.

Theorem 3.10.1. With the preceding notation, if u and v are composable cohomological correspondences, then

$$\mathrm{Tr}_{Z/S}((v \circ u)_*) = \mathrm{Tr}_{Y/S}(v_* \circ u_*).$$

This is the Lefschetz trace formula at the level of cohomological correspondences.

3.11–3.12. Interpreting the same construction in étale cohomology, cohomological correspondences act on ℓ -adic cohomology and recover the familiar fixed-point formulae. If X is a scheme over $\mathbb{F}_q$ and $F : X \rightarrow X$ is Frobenius, the associated correspondence gives the Grothendieck–Lefschetz trace formula.

4. Cup-products and trace formulae

4.0. We keep the preceding notation. Let X and Y be S -schemes, and let

$$L_1, L_2 \in \mathrm{Ob} D^-(X), \quad M_1, M_2 \in \mathrm{Ob} D^-(Y).$$

4.1. Cup-product pairings

Let $C \subset X \times_S Y$, with projections $p : C \rightarrow X$ and $q : C \rightarrow Y$. There are natural pairings

$$\mathrm{Hom}_C(p^* L_1, q^* M_1) \otimes \mathrm{Hom}_C(p^* L_2, q^* M_2) \longrightarrow \mathrm{Hom}_C(p^*(L_1 \otimes^L L_2), q^*(M_1 \otimes^L M_2)). \quad (4.1.1)$$

In particular, after specializing L_1 and M_2 to the structure object, one obtains

$$\mathrm{Hom}_C(p^* L, q^* M) \otimes \mathrm{Hom}_C(q^* M, p^* L) \longrightarrow \mathrm{Hom}_C(\mathcal{O}_C, \mathcal{O}_C) = H^0(C, \mathcal{O}_C). \quad (4.1.2)$$

There are analogous pairings for the internal derived Hom complexes:

$$\underline{\mathrm{RHom}}_C(p^*L_1, q^*M_1) \otimes^L \underline{\mathrm{RHom}}_C(p^*L_2, q^*M_2) \longrightarrow \underline{\mathrm{RHom}}_C(p^*(L_1 \otimes^L L_2), q^*(M_1 \otimes^L M_2)). \quad (4.1.4)$$

4.2. Local cohomology. Let $c : C \rightarrow X \times_S Y$ and $d : D \rightarrow X \times_S Y$ be morphisms, and let L, M, L', M' be complexes on the two factors. The cup-product in cohomology with supports has the form

$$H_C^i(X \times Y, p_1^*L \otimes^L p_2^*M) \otimes H_D^j(X \times Y, p_1^*L' \otimes^L p_2^*M') \longrightarrow H_{C \cap D}^{i+j}(\cdots). \quad (4.2.1)$$

For proper supports and correspondences

$$u \in \mathrm{Hom}(c^*p_1^*L, c^!p_2^*M), \quad v \in \mathrm{Hom}(d^*p_1^*L', d^!p_2^*M'),$$

one defines

$$\langle u, v \rangle \in \mathrm{Hom}(\mathrm{R}\Gamma_{C \cap D}(\mathcal{O}), K_X) \quad (4.2.3)$$

by the formula

$$\langle u, v \rangle = \mathrm{Tr}(u \cup v). \quad (4.2.4)$$

4.3. Projection formula

The projection formulae for these cup-products are obtained from the usual projection formula in the derived category, together with the compatibility of traces with adjunction. They express that pushing forward one factor before taking the pairing gives the same result as first taking the pairing and then applying the trace.

4.4. The principal formula

Theorem 4.4.1. Let X_i, Y_i ($i = 1, 2$) be S -schemes, and let $f_i : X_i \rightarrow Y_i$ be proper morphisms. Consider the cube 4.4.0 with cartesian horizontal faces and with c', c'', d', d'' proper. Under the stated tor-independence assumptions, and for $L_i \in \mathrm{Ob} D(X_i)_{\mathrm{parf}/S}$, the square

$$\begin{array}{ccc} f_*c'_*\underline{\mathrm{RHom}}(c'^*L_1, c'^!L_2) \otimes f_*c''_*\underline{\mathrm{RHom}}(c''^*L_2, c''^!L_1) & \xrightarrow{(1)} & f_*\underline{\mathrm{RHom}}(\mathrm{R}c'_*\mathcal{O}_{C'}, \mathcal{O}_X) \\ \downarrow & & \downarrow \\ d'_*\underline{\mathrm{RHom}}(d'^*M_1, d'^!M_2) \otimes d''_*\underline{\mathrm{RHom}}(d''^*M_2, d''^!M_1) & \xrightarrow{(3)} & \underline{\mathrm{RHom}}(\mathrm{R}d'_*\mathcal{O}_{D'}, \mathcal{O}_Y) \end{array}$$

is commutative. Here $M_i = f_{i*}L_i$; the arrow (3) is given by the cup-product 4.2.4; and (1) is obtained from the same construction after applying f_* and composing with the canonical

map from the tensor product of direct images to the direct image of a tensor product.

Corollary 4.5. The Lefschetz formula for cohomological correspondences follows from 4.4.1.

4.6. Trace calculation. For

$$u \in \mathrm{Hom}(c_1^* L_1, c_2^! L_2), \quad v \in \mathrm{Hom}(c_2^* L_2, c_1^! L_1),$$

one has

$$\mathrm{Tr}(f_* u, f_* v) = f_* \mathrm{Tr}(u, v). \quad (4.6.1)$$

4.7. Tensor products of correspondences. For $u_i \in \mathrm{Hom}_{X_i}(L_i, M_i)$ one defines

$$u_1 \boxtimes u_2 \in \mathrm{Hom}_{X_1 \times X_2}(L_1 \boxtimes L_2, M_1 \boxtimes M_2).$$

The trace is multiplicative:

$$\mathrm{Tr}(u_1 \boxtimes u_2) = \mathrm{Tr}(u_1) \mathrm{Tr}(u_2). \quad (4.7.1)$$

5. Complements

5.1. Duality and Hom-type functors

For a morphism $f : X \rightarrow Y$ of S -schemes and for $L, M \in \mathrm{Ob} D(X)$ put

$$\underline{\mathrm{RHom}}_f(L, M) = \underline{\mathrm{RHom}}(L, f^! M). \quad (5.1.1)$$

For $M \in \mathrm{Ob} D^-(Y)$ and $L \in \mathrm{Ob} D^+(X)$ there are adjunction isomorphisms

$$\mathrm{Hom}_X(L, f^! M) \xrightarrow{\sim} \mathrm{Hom}_Y(\mathrm{R}f_! L, M). \quad (5.1.2)$$

The associated Ext-type functors are

$$\mathrm{Ext}_f^i(L, M) = H^i(\underline{\mathrm{RHom}}_f(L, M)). \quad (5.2.1)$$

For a composite $X \xrightarrow{f} Y \xrightarrow{g} Z$ there are transitivity isomorphisms

$$\underline{\mathrm{RHom}}_{gf}(L, N) \xrightarrow{\sim} \underline{\mathrm{RHom}}_f(L, g^! N). \quad (5.3.1)$$

5.4. Base change

For a cartesian square

$$\begin{array}{ccc} X' & \xrightarrow{g'} & X \\ \downarrow f' & & \downarrow f \\ Y' & \xrightarrow{g} & Y, \end{array}$$

one has a canonical morphism

$$g^* \underline{\mathrm{R Hom}}_f(L, M) \circ \underline{\mathrm{R Hom}}_{f'}(g'^* L, g^* M), \quad (5.4.1)$$

which is an isomorphism when g is flat, or when L is perfect.

5.5–5.7. Künneth formulae

For X_1 and X_2 over S , and $L_i \in \mathrm{Ob} D^-(X_i)$, the Künneth morphism is

$$\mathrm{Rp}_{1*} L_1 \otimes_{\mathcal{O}_S}^L \mathrm{Rp}_{2*} L_2 \longrightarrow \mathrm{Rp}_*(L_1 \boxtimes L_2), \quad (5.5.1)$$

where $p_i : X_1 \times_S X_2 \rightarrow X_i$ and $p : X_1 \times_S X_2 \rightarrow S$. It is an isomorphism under tor-independence hypotheses. Similarly, for X_i, Y_i and $f_i : X_i \rightarrow Y_i$, one has

$$f_* \underline{\mathrm{R Hom}}(L_1, M_1) \otimes f_* \underline{\mathrm{R Hom}}(L_2, M_2) \circ f_* \underline{\mathrm{R Hom}}(L_1 \boxtimes L_2, M_1 \boxtimes M_2). \quad (5.6.1)$$

The Künneth morphisms are compatible with base change, composition, and the cup-product pairings.

5.8–5.9. Complements on correspondences

If

$$C \subset X \times_S Y, \quad D \subset Y \times_S Z,$$

then the composite correspondence is

$$D \circ C = \mathrm{pr}_{13}(C \times_Y D).$$

For composable cohomological correspondences u and v , the trace pairing satisfies

$$\langle v \circ u \rangle = \langle v, u \rangle, \quad (5.9.1)$$

in the sense of the pairings of 4.2.

6. Cohomological operations and the Lefschetz–Verdier formula

6.0. This section gives generalizations of the preceding constructions, leading to the Lefschetz–Verdier formula and to the Woods Hole formula.

6.1–6.2. Derived categories and relative duality

Let $f : X \rightarrow S$ be a morphism of schemes. Write $D(X)$ for the derived category of sheaves of $\mathcal{O}_X$ -modules, and $D(X)_{\text{parf}}$ for the full subcategory of perfect complexes. If f is proper and of finite tor-dimension, there is a functor

$$f^! : D^+(S) \rightarrow D^+(X) \quad (6.1.1)$$

right adjoint to Rf_* , together with a duality isomorphism

$$\underline{\text{R Hom}}_X(L, f^!M) \xrightarrow{\sim} \underline{\text{R Hom}}_S(Rf_*L, M). \quad (6.1.2)$$

The relative dualizing object is

$$K_{X/S} = f^!\mathcal{O}_S.$$

If f is smooth, purely of relative dimension n , then

$$K_{X/S} = \Omega_{X/S}^n[n].$$

For such f one has trace morphisms, for instance

$$\text{Tr}_f : R^{2n}f_*\Omega_{X/S}^n \rightarrow \mathcal{O}_S, \quad (6.2.1)$$

and more generally, for $K \in \text{Ob } D_{\text{parf}}^b(X)$,

$$\text{Tr}_f : Rf_*\underline{\text{R Hom}}(K, K_{X/S}) \rightarrow \mathcal{O}_S. \quad (6.2.2)$$

6.3. General Künneth formulae

In the situation of 1.6, suppose that X_1 and X_2 are tor-independent over S , and likewise Y_1 and Y_2 . If $L_i \in \text{Ob } D(X_i)_{\text{parf}/S}$, the Künneth arrow 1.6.4 is defined, using the fact that finite relative tor-dimension is preserved under direct image. It is an isomorphism: the verification is reduced by affine localization and a ech calculation to the affine case, where the assertion follows from tor-independence.

For $M_i \in \text{Ob } D(Y_i)_{\text{parf}/S}$, with $f_1^!M_1$ or $f_2^!M_2$ bounded, the Künneth arrow 1.7.3 is also

defined, and is an isomorphism when f_1 and f_2 are of finite tor-dimension. In particular, if X_1 and X_2 are of finite tor-dimension over S and tor-independent, 1.7.6 is an isomorphism.

One defines, as in 2.1, the complexes $\underline{\mathbf{R}}\underline{\mathbf{Hom}}(u, v)$ for pairs of arrows of type (2, 1) and of type (3, 4), assuming f proper in order to avoid the pro-objects that appear with $f_!$. Subject to the appropriate boundedness conditions, the evaluation and tensor-product arrows exist and behave as in the preceding sections.

6.4. Cohomological correspondences with support

Let X_1 and X_2 be S -schemes, and let $L_i \in \text{Ob } D(X_i)$. One considers four kinds of arrows over a morphism f :

$$\text{Hom}_f^{(1)}(L, M) = \text{Hom}(M, f_*L), \quad (6.4.1)$$

$$\text{Hom}_f^{(2)}(L, M) = \text{Hom}(f^*M, L), \quad (6.4.2)$$

$$\text{Hom}_f^{(3)}(L, M) = \text{Hom}(L, f^!M), \quad (6.4.3)$$

$$\text{Hom}_f^{(4)}(L, M) = \text{Hom}(f^!M, L). \quad (6.4.4)$$

Their elements are called arrows of type i from L to M over f , or f -arrows of type i . The partial adjunction formulae recover the usual descriptions; for instance, with suitable boundedness,

$$\text{Hom}_f^{(1)}(L, M) = \text{Hom}(M, f_*L),$$

and, for f proper,

$$\text{Hom}_f^{(3)}(L, M) = \text{Hom}(f_*L, M).$$

6.5. Direct images

Let $f_i : X_i \rightarrow Y_i$ ($i = 1, 2$) be proper S -morphisms, and put $f = f_1 \times_S f_2 : X \rightarrow Y$. Assume the finite tor-dimension and tor-independence hypotheses appearing in 2.4.0. If $L_i \in \text{Ob } D(X_i)_{\text{parf}/S}$, then, as in 3.3.1, there is an isomorphism

$$f_*\underline{\mathbf{R}}\underline{\mathbf{Hom}}_S(L_1, L_2) \xrightarrow{\sim} \underline{\mathbf{R}}\underline{\mathbf{Hom}}_S(f_{1*}L_1, f_{2*}L_2), \quad (6.5.1)$$

and hence, after applying $\text{R}\Gamma(Y, -)$,

$$f_* : \text{Hom}_S(L_1, L_2) \rightarrow \text{Hom}_S(f_{1*}L_1, f_{2*}L_2). \quad (6.5.2)$$

In the case $Y_1 = Y_2 = S$, this is again written $u \mapsto u_*$.

For a square of the form 3.7.1 with c proper and $K \in \text{Ob } D^+(C)$, there is a canonical arrow

$$f_* c_* c^! K \rightarrow d_* d^! f_* K, \quad (6.5.3)$$

which is defined as in 3.7.3. Taking $K = \underline{\text{R Hom}}_S(L_1, L_2)$ and using 6.5.1 gives

$$f_* c_* \underline{\text{R Hom}}(c_1^* L_1, c_2^! L_2) \circ d_* \underline{\text{R Hom}}(d_1^*(f_{1*} L_1), d_2^!(f_{2*} L_2)), \quad (6.5.4)$$

and on global Hom groups,

$$f_* : \text{Hom}(c_1^* L_1, c_2^! L_2) \circ \text{Hom}(d_1^*(f_{1*} L_1), d_2^!(f_{2*} L_2)). \quad (6.5.5)$$

These are the analogues of 3.7.5 and 3.7.6.

6.6. Cup-products and pairings

Let X_i be S -schemes of finite tor-dimension and tor-independent, and let $L_i \in \text{Ob } D(X_i)_{\text{parf}/S}$. The exterior tensor product of the evaluation morphisms

$$\underline{D}L_i \otimes L_i \rightarrow K_{X_i}$$

defines cup-product pairings analogous to 4.1.2–4.1.4.

Let $c : C \rightarrow X$ and $d : D \rightarrow X$ be proper morphisms, and let $e = c \times_X d : E \rightarrow X$. For $P, Q \in \text{Ob } D(X)_{\text{parf}/S}$ one has

$$c_* c^! P = \underline{\text{R Hom}}(\text{R}c_* \mathcal{O}_C, P),$$

and similarly for d . Assuming c and d of finite tor-dimension at the relevant points, the tensor product gives

$$c_* c^! P \otimes d_* d^! Q \rightarrow \underline{\text{R Hom}}(\text{R}c_* \mathcal{O}_C \otimes \text{R}d_* \mathcal{O}_D, P \otimes Q). \quad (6.6.1)$$

When c and d are tor-independent, the right-hand side is rewritten, by Künneth and 6.1.1, as $e_* e^!(P \otimes Q)$. Consequently one obtains pairings

$$\begin{aligned} c_* \underline{\text{R Hom}}(c_1^* L_1, c_2^! L_2) \otimes d_* \underline{\text{R Hom}}(d_2^* L_2, d_1^! L_1) \\ \longrightarrow \underline{\text{R Hom}}(\text{R}c_* \mathcal{O}_C \otimes \text{R}d_* \mathcal{O}_D, K_X), \end{aligned} \quad (6.6.2)$$

and, on global Hom groups,

$$\langle \ , \ \rangle : \text{Hom}(c_1^* L_1, c_2^! L_2) \otimes \text{Hom}(d_2^* L_2, d_1^! L_1) \circ \text{Hom}(\text{R}c_* \mathcal{O}_C \otimes \text{R}d_* \mathcal{O}_D, K_X). \quad (6.6.3)$$

These pairings are compatible with étale localization. If c and d are tor-independent, the

target is canonically e_*K_E , respectively $H^0(E, K_E)$.

6.7. Lefschetz–Verdier formula

Theorem 6.7. Let $f_i : X_i \rightarrow Y_i$ ($i = 1, 2$) be proper S -morphisms, put $f = f_1 \times_S f_2 : X \rightarrow Y$, and let $p_i : X \rightarrow X_i$, $q_i : Y \rightarrow Y_i$ be the projections. Suppose given a cube as in 4.4.0, with cartesian horizontal faces, with c' , c'' , d' , d'' proper, and with the required finite tor-dimension assumptions along the supports. Let $L_i \in \text{Ob } D(X_i)_{\text{parf}/S}$ and suppose the pairs (X_1, X_2) , (X_1, Y_2) , (Y_1, X_2) , and (Y_1, Y_2) are tor-independent over S , while X_i and Y_i have finite tor-dimension over S . Then the square

$$\begin{array}{ccc} f_*c'_*\underline{\text{R Hom}}(c_1'^*L_1, c_2'^!L_2) \otimes f_*c''_*\underline{\text{R Hom}}(c_2''^*L_2, c_1''^!L_1) & \rightarrow & f_*\underline{\text{R Hom}}(\text{R}c'_*\mathcal{O}_{C'} \otimes \text{R}c''_*\mathcal{O}_{C''}, K_X) \\ \downarrow & & \downarrow \\ d'_*\underline{\text{R Hom}}(d_1'^*M_1, d_2'^!M_2) \otimes d''_*\underline{\text{R Hom}}(d_2''^*M_2, d_1''^!M_1) & \rightarrow & \underline{\text{R Hom}}(\text{R}d'_*\mathcal{O}_{D'} \otimes \text{R}d''_*\mathcal{O}_{D''}, K_Y) \end{array}$$

is commutative, where $M_i = f_{i*}L_i$. The horizontal arrows are the cup-product pairings of 6.6.2; the left vertical arrow is the tensor product of the direct-image maps 6.5.4; the right vertical arrow is obtained from relative duality and the canonical base-change morphisms.

The proof is the same as that of 4.4, with the necessary substitutions in the duality formalism. If c' and c'' are tor-independent, and likewise d' and d'' , the right vertical arrow can be read simply as

$$f_*c_*K_C \rightarrow d_*K_D,$$

given by the adjunction map $g_*K_C = g_*g^!K_D \rightarrow K_D$. The general statement is kept because it includes excess-intersection cases.

As a consequence, under the hypotheses of 6.7, for

$$u \in \text{Hom}(c_1'^*L_1, c_2'^!L_2), \quad v \in \text{Hom}(c_2''^*L_2, c_1''^!L_1),$$

one has

$$\langle f_*u, f_*v \rangle = f_*\langle u, v \rangle, \tag{6.7.2}$$

where f_*u and f_*v are direct-image correspondences in the sense of 6.5.5, and where the f_* on the right is the morphism

$$\text{Hom}(\text{R}c'_*\mathcal{O}_{C'} \otimes \text{R}c''_*\mathcal{O}_{C''}, K_X) \circ \text{Hom}(\text{R}d'_*\mathcal{O}_{D'} \otimes \text{R}d''_*\mathcal{O}_{D''}, K_Y). \tag{6.7.3}$$

When $Y_1 = Y_2 = S = D' = D''$, formula (6.7.2) is the Lefschetz–Verdier formula.

6.8. The Hodge cohomology class

Let $f : X \rightarrow S$ be smooth, purely of relative dimension N , and let $i : Y \hookrightarrow X$ be a regular immersion of codimension d ; equivalently, Y is locally cut out by a regular sequence of d equations, or is a local complete intersection of codimension d . Put $g = fi$. One attaches to Y an element

$$\mathrm{cl}_{X/S}(Y) \in \mathrm{Ext}_{\mathcal{O}_X}^d(\mathcal{O}_Y, \Omega_{X/S}^d), \quad (6.8.1)$$

called the *cohomology class associated with Y* .

By the purity theorem, one has

$$\underline{\mathrm{Ext}}_{\mathcal{O}_X}^i(\mathcal{O}_Y, \Omega_{X/S}^d) = \begin{cases} 0, & i \neq d, \\ \underline{\mathrm{Hom}}(\Lambda^d N_{Y/X}, \Omega_{X/S}^d \otimes \mathcal{O}_Y), & i = d, \end{cases} \quad (6.8.2)$$

where $N_{Y/X} = I/I^2$ is the conormal sheaf, $I = \mathrm{Ker}(\mathcal{O}_X \rightarrow \mathcal{O}_Y)$. Therefore

$$\mathrm{Ext}_{\mathcal{O}_X}^d(\mathcal{O}_Y, \Omega_{X/S}^d) = H^0(Y, \underline{\mathrm{Hom}}(\Lambda^d N_{Y/X}, \Omega_{X/S}^d \otimes \mathcal{O}_Y)). \quad (6.8.3)$$

If Y is defined by the ideal generated by a regular sequence $t_1, \dots, t_d$, then $\Lambda^d N_{Y/X}$ is generated by $\bar{t}_1 \wedge \dots \wedge \bar{t}_d$, and $\mathrm{cl}_{X/S}(Y)$ corresponds to

$$\bar{t}_1 \wedge \dots \wedge \bar{t}_d \longmapsto dt_1 \wedge \dots \wedge dt_d. \quad (6.8.5)$$

It is multiplicative for transversal intersections:

$$\mathrm{cl}_{X/S}(Y \cap Z) = \mathrm{cl}_{X/S}(Y) \mathrm{cl}_{X/S}(Z). \quad (6.8.6)$$

If Y has codimension equal to the relative dimension and is finite flat over S , the trace is

$$g_* g^* F = F \otimes g_* \mathcal{O}_Y \rightarrow F, \quad (6.8.7)$$

induced by $\mathrm{Tr}_{Y/S} : g_* \mathcal{O}_Y \rightarrow \mathcal{O}_S$.

6.9–6.10. Woods Hole formula

Let X and Y be proper smooth S -schemes, of relative dimensions m and n , and set $Z = X \times_S Y$. Let

$$a : A \hookrightarrow Z, \quad b : B \hookrightarrow Z$$

be regular immersions such that $a_2 : A \rightarrow Y$ and $b_1 : B \rightarrow X$ are finite and flat. Write $p_1 : Z \rightarrow X$, $p_2 : Z \rightarrow Y$, $a_i = p_i a$, $b_i = p_i b$, and put $C = A \times_Z B$. Assume C is finite flat over S . Let $L \in \mathrm{Ob} D(X)_{\mathrm{parf}}$, $M \in \mathrm{Ob} D(Y)_{\mathrm{parf}}$.

For every r one has the decomposition

$$\Omega_{Z/S}^r = \bigoplus_{i+j=r} \Omega_{Z/Y}^i \otimes \Omega_{Z/X}^j. \quad (6.9.0)$$

The identifications of $p_2^!M$ and $p_1^!L$ give

$$\mathrm{Hom}(a_1^*L, a_2^!M) = H^0\left(A, \underline{\mathrm{RHom}}(a_1^*L, a_2^*M) \otimes \Lambda^m N_{A/Z}^\vee \otimes \Omega_{Z/Y}^m\right), \quad (6.9.1)$$

and similarly

$$\mathrm{Hom}(b_2^*M, b_1^!L) = H^0\left(B, \underline{\mathrm{RHom}}(b_2^*M, b_1^*L) \otimes \Lambda^n N_{B/Z}^\vee \otimes \Omega_{Z/X}^n\right). \quad (6.9.2)$$

Let $u \in \mathrm{Hom}(a_1^*L, a_2^*M)$ and $v \in \mathrm{Hom}(b_2^*M, b_1^*L)$. On C define

$$\langle u, v \rangle = \mathrm{Tr}(u_C v_C) = \mathrm{Tr}(v_C u_C) \in H^0(C, \mathcal{O}_C). \quad (6.9.3)$$

The cohomological correspondences $u^!$ and $v^!$ are obtained from u and v by composition with the trace morphisms, equivalently by multiplication with $\mathrm{cl}_{Z/Y}(A)$ and $\mathrm{cl}_{Z/X}(B)$. Their cup-product is

$$\langle u^!, v^! \rangle = \langle u, v \rangle \mathrm{cl}_{Z/Y}(A) \mathrm{cl}_{Z/X}(B). \quad (6.9.4)$$

The Lefschetz–Verdier formula gives

$$\langle (u^!)_*, (v^!)_* \rangle = \mathrm{Tr}_{Z/S}\left(\langle u, v \rangle \mathrm{cl}_{Z/Y}(A) \mathrm{cl}_{Z/X}(B)\right). \quad (6.9.5)$$

Thus:

Theorem 6.10. Under the hypotheses of 6.9,

$$\mathrm{Tr}\left((u^!)_* \circ (v^!)_*\right) = \mathrm{Tr}_{Z/S}\left(\langle u, v \rangle \mathrm{cl}_{Z/Y}(A) \mathrm{cl}_{Z/X}(B)\right).$$

For $L = \mathcal{O}_X$, $M = \mathcal{O}_Y$, and u, v the identity maps, this recovers the classical intersection-number formula. The same formalism admits variants for other types of correspondences, singular schemes, continuous coefficients, and the usual applications to étale trace formulae and congruence formulae.

Exposé III B. Calculation of local terms

by Luc Illusie

0. Introduction

The purpose of this exposé is to calculate the local terms occurring in the Lefschetz–Verdier trace formula for several kinds of correspondences. The results generalize those of Exposé III A and complete the formulae of SGA 5, Exposé III.

Let X be a scheme of finite type over an algebraically closed field k of characteristic p . One considers cohomological correspondences

$$u : c_1^* L \rightarrow c_2^! L, \quad c = (c_1, c_2) : C \rightarrow X \times X.$$

The local term of u at an isolated fixed point $x \in X$ is defined as the trace of the endomorphism induced by u on the appropriate local complex. The plan is as follows. Section 1 treats correspondences in general position between curves. Section 2 reduces the proof to an assertion of vanishing. Section 3 reduces to the tame case. Section 4 finishes the proof of the main theorem. The second part treats equivariant correspondences and non-commutative traces.

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I. Correspondences in general position between curves

Throughout this part, $S = \operatorname{Spec}(k)$ with k algebraically closed of characteristic p , and Λ is a commutative noetherian ring killed by an integer prime to p . For every S -scheme X , let $D(X)$ be the derived category of sheaves of Λ -modules on X , and let $D_{\text{ctf}}(X)$ be the full subcategory formed by complexes of finite tor-dimension with constructible cohomology.

1. Statement of the theorem and corollaries

Let X and Y be the strict localizations at closed points of smooth separated S -curves. Let A and B be strictly local S -schemes, and let

$$a : A \rightarrow X \times_S Y, \quad b : B \rightarrow X \times_S Y$$

be S -morphisms. Write p_1, p_2 for the projections, set $a_i = p_i a$, $b_i = p_i b$, and denote by x and y the closed points of X and Y . Assume that a_2 and b_1 are finite and surjective, and that

$$A \times_{X \times_S Y} B$$

is reduced to the closed point z mapping to (x, y) .

Let

$$L \in \text{Ob } D_{\text{ctf}}(X), \quad M \in \text{Ob } D_{\text{ctf}}(Y),$$

and let

$$u \in \text{Hom}(a_1^* L, a_2^! M), \quad v \in \text{Hom}(b_2^* M, b_1^! L).$$

The induced map

$$u_z : L_x = \text{R}\Gamma(X, L) \rightarrow \text{R}\Gamma(Y, M) = M_y$$

is the composite

$$L_x \xrightarrow{a_1^*} \text{R}\Gamma(A, a_1^* L) \xrightarrow{\text{R}\Gamma(A, u)} \text{R}\Gamma(A, a_2^! M) \xrightarrow{a_{2*}} \text{R}\Gamma(Y, M),$$

where a_{2*} is defined by the adjunction map $a_{2*} a_2^! M \rightarrow M$. The morphism $v_z : M_y \rightarrow L_x$ is defined similarly. Since L_x and M_y are perfect complexes of Λ -modules, one can form

$$\langle u_z, v_z \rangle = \text{Tr}(u_z v_z) = \text{Tr}(v_z u_z) \in \Lambda. \quad (1.1.1)$$

Put $K_X = \Lambda_X(1)[2]$ and $K_Y = \Lambda_Y(1)[2]$. By duality, the functors

$$D_X = \text{R Hom}(-, K_X), \quad D_Y = \text{R Hom}(-, K_Y)$$

are dualizing on $D_{\text{ctf}}(X)$ and $D_{\text{ctf}}(Y)$. As in III 4.1.2 and III 4.2.3, the cohomological correspondences u and v have a cup-product local term

$$\langle u, v \rangle \in \Lambda. \quad (1.1.2)$$

If the data are obtained by strict localization from global curves X', Y', A', B' , this is the local term of III 4.2.7 at the isolated point z' .

In general the two quantities (1.1.1) and (1.1.2) are not equal. The main theorem of the first part gives a sufficient general-position condition.

Theorem 1.2. Under the hypotheses of 1.1, assume in addition that A and B are in general position: the intersection of the tangent cones at (x, y) to their images in $X \times_S Y$ is reduced to (x, y) , and similarly A and $X \times \{y\}$ are in general position, as are $\{x\} \times Y$ and B . Then

$$\langle u_z, v_z \rangle = \langle u, v \rangle. \quad (1.2.1)$$

Corollary 1.3. Let F be an endomorphism of a proper smooth curve X/S such that the fixed-point scheme X^F is finite and consists of points at which the graph of F is transverse to the diagonal. If $L \in \text{Ob } D_{\text{ctf}}(X)$ and $u \in \text{Hom}(F^* L, L)$, then

$$\text{Tr}((F, u), \text{R}\Gamma(X, L)) = \sum_{x \in X^F} \text{Tr}(u_x, L_x).$$

For L concentrated in degree 0 this is the Artin–Verdier result. Passing to the limit gives the usual ℓ -adic form.

Corollary 1.3.1. With X and F as above, let L be a constructible $\bar{\mathbb{Q}}_\ell$ -sheaf on X , with $\ell \neq p$, and let $u \in \text{Hom}(F^*L, L)$. Then

$$\sum_i (-1)^i \text{Tr}((F, u), H_c^i(\bar{X}, L)) = \sum_{x \in X^F} \text{Tr}(u_x, L_x).$$

The hypotheses apply in particular when (F, u) is a Frobenius correspondence; one obtains Grothendieck’s trace formula.

There is also a dual form. If b_2 and a_1 are finite and surjective, consider the dual correspondences

$$Du \in \text{Hom}(a_2^*DM, a_1^!DL), \quad Dv \in \text{Hom}(b_1^*DL, b_2^!DM).$$

Under the corresponding general-position assumptions, applying Theorem 1.2 to Du and Dv yields

$$\langle u, v \rangle = \langle (Du)_z, (Dv)_z \rangle. \quad (1.4.1)$$

Corollary 1.5. Let X and Y be proper smooth curves over S , and let $a : A \rightarrow X \times_S Y$, $b : B \rightarrow X \times_S Y$ be correspondences with A and B proper curves over S . Let $L \in \text{Ob } D_{\text{ctf}}(X)$, $M \in \text{Ob } D_{\text{ctf}}(Y)$, and let u and v be cohomological correspondences as above. Assume $A \times_{X \times_S Y} B$ is finite over S and decomposes as $C = C_d \cup C_t$, where the points of C_d satisfy the general-position hypotheses for (a_2, b_1) and the points of C_t satisfy the dual general-position hypotheses. If $\underline{u}$ and $\underline{v}$ are the induced morphisms on cohomology, then

$$\langle \underline{u}, \underline{v} \rangle = \sum_{z \in C_d} \langle u_z, v_z \rangle + \sum_{z \in C_t} \langle (Du)_z, (Dv)_z \rangle. \quad (1.5.1)$$

The ℓ -adic limit form of this corollary contains Langlands’ Proposition 7.12 as a special case. For the identity coefficients, taking u and v to be the fundamental classes of A and B , Theorem 1.2 reads

$$\deg(a_2) \deg(b_1) = \langle \text{cl}(A), \text{cl}(B) \rangle, \quad (1.7.1)$$

which is the intersection multiplicity of the direct-image cycles under the stated general-position hypotheses.

2. Reduction to a vanishing theorem

Let $U = X - \{x\}$ and $V = Y - \{y\}$, with inclusions $i : U \hookrightarrow X$ and $j : V \hookrightarrow Y$. Since a_2 and b_1 are finite, $a_2^{-1}(y) = s$ and $b_1^{-1}(x) = t$, so the inverse images of U and V fit into cartesian diagrams.

Lemma 2.2. To prove Theorem 1.2 it suffices to treat the corresponding graded cases obtained from the filtrations

$$F^n L = \begin{cases} L, & n < 0, \\ i_! i^* L, & n = 1, \\ 0, & n > 1, \end{cases} \quad F^n M = \begin{cases} M, & n < 0, \\ j_! j^* M, & n = 1, \\ 0, & n > 1. \end{cases}$$

The morphisms u and v are morphisms of filtered complexes and, by additivity of cup-products,

$$\langle u, v \rangle = \langle \mathrm{gr}^0 u, \mathrm{gr}^0 v \rangle + \langle \mathrm{gr}^1 u, \mathrm{gr}^1 v \rangle.$$

The graded pieces reduce the theorem to the cases supported at the closed points and on their complements.

Lemma 2.3. Under the hypotheses of 1.1, if $L|_U = 0$ and $M|_V = 0$, then

$$\langle u_z, v_z \rangle = \langle u, v \rangle.$$

The assertion is a direct form of the Lefschetz formula for the projections $\{x\} \rightarrow S$ and $\{y\} \rightarrow S$.

The remaining case is reduced to a vanishing statement. After replacing the data by global curves and by constructible bounded complexes, one is led to prove the vanishing of a restriction morphism of the form

$$\mathrm{R}\Gamma(T - A, i_! E) \rightarrow \mathrm{R}\Gamma(B - t, i_! E|_{B-t})$$

for a two-dimensional strict local scheme T , with divisors A , B , and X meeting in the prescribed general position. This is the proposition below, after the standard reductions on coefficients.

3. Reduction to the tame case

The proof first reduces to the case where Λ is killed by a power of a prime $\ell \neq p$ and where the constructible sheaves in question are locally free with finite monodromy through finite ℓ -groups. The reductions use dévissage, passage to finite étale covers, and the behavior of the fundamental group under strict localization. The geometrical part is to show that the general-position hypotheses are preserved by the finite covers used in this reduction.

Let $\tilde{A}_i$ and $\tilde{B}_j$ be the normalizations of the branches of A and B at the closed point. The tangent directions of the branches are represented by non-zero elements λ_i and μ_j of the residue field. General position is the condition

$$\lambda_i \neq 0, \quad \mu_j \neq 0, \quad \lambda_i \neq \mu_j \quad \text{for all } i, j. \quad (3.3.1)$$

After passing to the finite covers, the new tangent directions λ'_i and μ'_j satisfy

$$\lambda'_i = C\lambda_i, \quad \mu'_j = C\mu_j$$

for a single $C \in k^*$; hence the inequalities (3.3.1) are preserved.

4. End of the proof of Theorem 1.2

It remains to establish the required vanishing under the coefficient hypotheses just stated. The assertion is the following.

Proposition 4.2. Let T be the strict localization at a closed point of a smooth two-dimensional S -scheme, with closed point t . Let A , B , and X be closed one-dimensional subschemes of T . Let

$$i : T - (A \cup X) \hookrightarrow T - A$$

be the inclusion. Assume X is regular, and that A is in general position with respect to B and X ; that is, the intersection of the tangent cone of A at t with the tangent cone of $B \cup X$ is reduced to t . Let $E \in \text{Ob } D^b(T - (A \cup X))$ have constructible locally constant cohomology, with monodromy factoring through an ℓ -group. Then the restriction map

$$R\Gamma(T - A, i_! E) \rightarrow R\Gamma(B - t, i_! E|_{B-t})$$

is zero.

The proof rests on the following lemma.

Lemma 4.3. Let Z be a connected regular noetherian scheme of dimension 2, with residual characteristics prime to ℓ . Let X and Y be regular divisors meeting transversely at a point z , giving a diagram of inclusions

$$\begin{array}{ccc} Z - X & \xrightarrow{j} & Z \\ \uparrow m & & \uparrow i \\ Y - z & \xrightarrow{q} & Y. \end{array}$$

Assume purity:

$$R^q j_*(\mathbb{Z}/\ell) = 0 \quad (q > 1).$$

Let $L \in \text{Ob } D^b(Z - (X \cup Y), \Lambda)$ have constructible locally constant cohomology with monodromy through an ℓ -group. Then:

- (a) the natural map $L \rightarrow m^* Rj_* L$ is an isomorphism, and $m^* Rj_* L$ has constructible locally constant cohomology;

(b) if Y is connected and $\bar{s}$ is a geometric point of Y localized at a closed point $s \neq z$, then

$$\mathrm{R}\Gamma(Y, m^* Rj_* L) o (Rj_* L)_{\bar{s}}$$

is zero;

(c) under the hypotheses of (b), if C is a one-dimensional closed subscheme of Z with $Y \cap C = s$, and T is the strict localization of C at s , then

$$\mathrm{R}\Gamma(Z - Y, i_! L) o \mathrm{R}\Gamma(T - s, i_! L|_{T-s})$$

is zero.

Proof of Lemma 4.3. The localization triangle gives an isomorphism

$$i^* Rj_* i_! L \xrightarrow{\sim} Rq_* q^* i_! L = Rq_* L.$$

By adjunction this yields $i_! L \rightarrow m_* m^* Rj_* L$, an isomorphism away from z . It remains to check that the stalk at $\bar{z}$ of $Rj_* i_! L$ is zero. By dévissage one reduces to $L = \mathbb{Z}/\ell$ with trivial action. Then the assertion is a compact-support calculation: in degrees < 1 it follows from Abhyankar's lemma, and in degrees > 1 from purity.

For (b), choose specialization maps from a geometric generic point of Y to the strict localizations at z and s . The resulting square of stalks shows that the restriction morphism factors through the stalk at z , which is zero. Part (c) follows by applying (b) to the commutative square of restriction maps from $Z - Y$ to the punctured strict localization of C at s .

Proof of Proposition 4.2. Let $\varepsilon : T' \rightarrow T$ be the blow-up of the maximal ideal of T , and let $D = \varepsilon^{-1}(t)$ be the exceptional divisor. Let A' , B' , and X' be the pure transforms of A , B , and X . Then T' is regular, connected, and two-dimensional; X' is a regular divisor meeting D transversely in one point; and the general-position hypotheses mean that A' is disjoint from $B' \cup X'$.

Pull E back to $T' - (A' \cup D \cup X')$. It suffices to prove the vanishing of the restriction

$$\mathrm{R}\Gamma(T' - (A' \cup D), i_! E') o \mathrm{R}\Gamma(B' - (B' \cap D), i_! E'|_{B' - (B' \cap D)}).$$

The scheme B' is a disjoint union of strict local one-dimensional schemes B'_j , each meeting D in a single point b_j . Apply Lemma 4.3(c) to

$$Z = T' - A', \quad X = X', \quad Y = D - (D \cap A'), \quad C = B'_j,$$

using the relative purity theorem to verify the purity hypothesis. Each resulting restriction map is zero, and therefore the displayed restriction is zero. This proves Proposition 4.2, hence the required vanishing, and so Theorem 1.2.

II. Equivariant correspondences

5. Non-commutative traces

Fix, for this section, a topos T and a commutative unital ring K of T . The rings considered below are associative and unital, but not necessarily commutative. A K -algebra means an associative unital ring A of T together with a unital homomorphism $K \rightarrow A$ whose image is contained in the center of A .

5.0. Introduction

Let E be a K -module which is locally a direct factor of a finite free module. There is then a local trace homomorphism

$$\mathrm{Tr}_K : \mathrm{Hom}_K(E, E) \rightarrow K, \quad (5.0.1)$$

defined by composing the evaluation map

$$\mathrm{Hom}_K(E, K) \otimes E \rightarrow K, \quad f \otimes x \mapsto f(x), \quad (5.0.2)$$

with the inverse of the canonical tensor-product isomorphism

$$\mathrm{Hom}_K(E, K) \otimes E \xrightarrow{\sim} \mathrm{Hom}_K(E, E), \quad f \otimes y \mapsto (x \mapsto f(x)y). \quad (5.0.3)$$

For any complex E of K -modules the same two maps induce morphisms of complexes

$$\mathrm{Hom}_K^\bullet(E, K) \otimes E \rightarrow K, \quad (5.0.4)$$

$$\mathrm{Hom}_K^\bullet(E, K) \otimes E \rightarrow \mathrm{Hom}_K^\bullet(E, E). \quad (5.0.5)$$

After derivation, for $E \in \mathrm{Ob} D^-(K)$, these give

$$\mathrm{R Hom}_K(E, K) \otimes_K^L E \rightarrow K, \quad (5.0.6)$$

$$\mathrm{R Hom}_K(E, K) \otimes_K^L E \rightarrow \mathrm{R Hom}_K(E, E). \quad (5.0.7)$$

If E is perfect, the second map is an isomorphism, and composing its inverse with the first gives the global trace

$$\mathrm{Tr}_K : \mathrm{R Hom}_K(E, E) \rightarrow K. \quad (5.0.8)$$

The point is to examine to what extent K may be replaced by a K -algebra A , especially the algebra of a finite group. The analogue of (5.0.3) causes no difficulty; the evaluation map does. If E is a left A -module, the map

$$\mathrm{Hom}_A(E, A) \otimes E \rightarrow A, \quad f \otimes x \mapsto f(x),$$

does not in general factor through $\mathrm{Hom}_A(E, A) \otimes_A E$. It factors, after quotienting, as

$$\mathrm{Hom}_A(E, A) \otimes_A E \rightarrow A_K, \quad (5.0.9)$$

where A_K is the quotient of A by the K -submodule locally generated by the commutators $ab - ba$. If E is locally a direct factor of a finite free left A -module, then

$$\mathrm{Hom}_A(E, A) \otimes_A E \xrightarrow{\sim} \mathrm{Hom}_A(E, E), \quad (5.0.10)$$

and hence one obtains

$$\mathrm{Tr}_A : \mathrm{Hom}_A(E, E) \rightarrow A_K. \quad (5.0.11)$$

For $A = K$ this is (5.0.1). When T is the punctual topos, it agrees with the trace of Stallings and Grothendieck. The rest of the section derives these maps for complexes and studies their behavior under change of scalars and under exterior tensor products; the resulting constructions will be used in 6 to define non-commutative Lefschetz–Verdier local terms.

5.1. Notation

For a K -algebra A , let A° be the opposite algebra and let

$$A^e = A \otimes_K A^\circ$$

be its enveloping algebra. Giving an A -bimodule is the same as giving a left, or right, A^e -module. If E is a bimodule, recall that

$$H_0(A, E) = A \otimes_{A^e} E = E/[A, E]$$

is the quotient of E by the submodule generated locally by elements $ax - xa$. We shall sometimes write E_A for this quotient. Thus A_K is the quotient introduced in (5.0.9). If $A = K[G]$, where G is an ordinary finite group, then A_K is the free K -module on the set of conjugacy classes of G .

For K -algebras A and B we use the Cartan–Eilenberg notation ${}_A E$, E_B , and ${}_A E_B$ for left A -modules, right B -modules, and A - B -bimodules. We write $D(A)$ for the derived category of left A -modules and $D(A, B)$ for the derived category of A - B -bimodules.

5.2–5.4. Adjunction, evaluation, and tensor-product maps

For K -algebras A and B , in the situation $({}_A E, {}_B F_A, {}_B G)$, there is the canonical adjunction

$$\mathrm{Hom}_A(E, \mathrm{Hom}_B(F, G)) \xrightarrow{\sim} \mathrm{Hom}_B(F \otimes_A E, G), \quad f \mapsto (y \otimes x \mapsto f(x)(y)). \quad (5.2.1)$$

It derives to

$$\mathrm{R Hom}_A(E, \mathrm{R Hom}_B(F, G)) \xrightarrow{\sim} \mathrm{R Hom}_B(F \otimes_A^L E, G), \quad (5.2.2)$$

for $E \in \mathrm{Ob } D^-(A)$, $F \in \mathrm{Ob } D^-(B, A^\circ)$, and $G \in \mathrm{Ob } D^+(B)$. One also obtains

$$\mathrm{R Hom}_{A^e}(E, \mathrm{R Hom}_B(F, G)) \xrightarrow{\sim} \mathrm{R Hom}_B(F \otimes_A^L E, G). \quad (5.2.3)$$

For a K -algebra A and data $({}_A E, {}_{A^\circ} F)$, there is an isomorphism

$$E \otimes_{A^e} F \xrightarrow{\sim} (E \otimes_K F) \otimes_{A^e} A. \quad (5.3.1)$$

For left A -modules E, F , the evaluation map

$$\mathrm{Hom}_A(E, F) \otimes_A E \rightarrow F, \quad f \otimes x \mapsto f(x), \quad (5.3.2)$$

is A^e -linear after passing to coinvariants, and induces

$$\mathrm{Hom}_A(E, F) \otimes_{A^e} E \rightarrow F \otimes_{A^e} A = F_A. \quad (5.3.3)$$

If A is flat over K , this derives to

$$E \otimes_{A^e}^L F \xrightarrow{\sim} (E \otimes_K^L F) \otimes_{A^e}^L A, \quad (5.3.4)$$

and to the derived evaluation map

$$\mathrm{R Hom}_A(E, F) \otimes_{A^e}^L E \rightarrow F \otimes_{A^e}^L A. \quad (5.3.5)$$

There is also a canonical tensor-product map, for $({}_A E, {}_A F_{A^\circ}, {}_A G_A)$,

$$\mathrm{Hom}_A(E, F) \otimes_A G \rightarrow \mathrm{Hom}_A(E, F \otimes_A G), \quad u \otimes y \mapsto (x \mapsto u(x) \otimes y). \quad (5.4.1)$$

Its derived form is

$$\mathrm{R Hom}_A(E, F) \otimes_A^L G \rightarrow \mathrm{R Hom}_A(E, F \otimes_A^L G). \quad (5.4.2)$$

This morphism is not always an isomorphism, but it is one when E is perfect. The examples intended for later use are $F = A$, variants of A twisted by an endomorphism, and $F = M \otimes_K A^\circ$ with M a dualizing complex.

5.5–5.7. Filtered and derived non-commutative traces

Let $DF(A)$ be the derived category of finite filtered complexes of left A -modules. The preceding adjunction, evaluation, and tensor-product morphisms have filtered derived analogues, compatible with passage to associated graded objects.

Assume A flat over K , and let $D(A)_{\mathrm{parf}}$ be the subcategory of complexes of finite perfect

amplitude. For $E \in \text{Ob } D(A)_{\text{parf}}$ and $F \in \text{Ob } D^-(A^\circ)$, define

$$\text{Tr}_A : \text{R Hom}_A(E, F \otimes_A^L E) \circ F \otimes_{A^e}^L A \quad (5.6.1)$$

by composing the inverse of (5.4.2), for $G = E$, with the evaluation map (5.3.5). On H^0 this gives

$$\text{Tr}_A : \text{Hom}_A(E, F \otimes_A E) \circ H^0(F)_A. \quad (5.6.2)$$

For $F = A$ one obtains

$$\text{Tr}_A : \text{R Hom}_A(E, E) \rightarrow A_K, \quad (5.6.3)$$

$$\text{Tr}_A : \text{Hom}_A(E, E) \rightarrow H^0(T, A_K). \quad (5.6.4)$$

The second map is local and elementary; the first is the global derived trace.

For filtered perfect complexes, the trace is compatible with the associated graded object. If F is concentrated in filtration degree 0 and u is a morphism in $DF(A)$, then

$$\text{Tr}_A(u) = \sum_i \text{Tr}_{A, \text{gr}}(\text{gr}^i u). \quad (5.6.5)$$

In particular, for $u : E \rightarrow F \otimes_A^L E$,

$$\text{Tr}(u[1]) = -\text{Tr}(u), \quad (5.6.6)$$

where $u[1]$ is the shifted morphism.

These traces simultaneously generalize the usual commutative traces of SGA 6 and the non-commutative traces of Stallings and Grothendieck. For $A = K$ they reduce to the standard trace maps. Moreover, for $u : E \rightarrow F$ and $v : F \rightarrow E$ between perfect A -complexes,

$$\text{Tr}_{A_K}(u \otimes v) = \text{Tr}_A(vu) \in A_K. \quad (5.7.1)$$

Homotopic endomorphisms have the same trace, because (5.6.4) is induced by a morphism of complexes.

5.8–5.10. Composition and restriction of scalars

For perfect data $({}_A E, {}_B F_A, {}_B G)$, the trace maps are compatible with composition: there is a commutative square

$$\begin{array}{ccc} \text{R Hom}_A(E, F \otimes_A E) \otimes \text{R Hom}_B(F, G \otimes_B F) & \xrightarrow{\circ} & \text{R Hom}_A(E, G \otimes_B F \otimes_A E) \\ \downarrow \text{Tr}_A \otimes \text{Tr}_B & & \downarrow \text{Tr}_B \\ F_A \otimes G_B & \longrightarrow & (G \otimes_B F)_A. \end{array}$$

The analogous square with A and B interchanged is also commutative.

Let $A \rightarrow B$ be a morphism of K -algebras with B flat over A . For $E \in \text{Ob } D^-(B)$, forgetting scalars gives

$$\text{R Hom}_B(E, E) \rightarrow \text{R Hom}_A(E, E). \quad (5.10.1)$$

Together with (5.6.3) for B and for A , this yields the commutative diagram

$$\begin{array}{ccc} \text{R Hom}_B(E, E) & \rightarrow & \text{R Hom}_A(E, E) \\ \downarrow & & \downarrow \\ B_K & \rightarrow & A_K. \end{array}$$

Equivalently, for $u \in \text{Hom}_B(E, E)$,

$$\text{Tr}_A(u) = \text{Tr}_{B/A}(\text{Tr}_B(u)), \quad (5.10.3)$$

where $\text{Tr}_{B/A} : B_K \rightarrow A_K$ is the trace associated with restriction of scalars.